No Kid Is an Island: Peer Effects and Intergenerational Mobility

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November 5, 2026

Abstract

How much, and how, do social interactions shape intergenerational mobility? Exploiting within-school variation in schoolmates' parental backgrounds among Danish high-school students, I show that a $1 increase in average peers' parental earnings raises annual adult earnings by $0.10. Combined with limited sorting between Danish high schools, this effect drives 3.6% of the difference in adult earnings between children from the top and bottom tercile of the parental earnings distribution. Increasing sorting to the rate observed among U.S. census tracts, the same contribution would be 11.2%. Employment trajectories reveal a key mechanism: as former schoolmates' careers advance, they facilitate access to higher-paying firms. A formal mediation analysis exploiting peers' promotion timing shows that improved access to higher-paying jobs accounts for 13% of the returns to exposure. Peer exposure emerges as a determinant of intergenerational mobility, shaping *access* to opportunities besides individual productivity.

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I thank Giacomo De Giorgi, Enrico Moretti and Frederic Robert-Nicoud for precious guidance and support. Gratitude is also extended to Ainoa Aparicio-Fenoll, Pietro Biroli, Sydnee Caldwell, Thomas Cornelissen, Christian Dustmann, Albrecht Glitz, Hilary Hoynes, Patrick Kline, Joan Llull, Rasmus Landersø, Valentina Melenteyeva, Michele Pellizzari, Ricardo Perez-Truglia, Jörn-Steffen Pischke, Luigi Pistaferri, Pauline Rossi, Kjell Salvanes, Jan Stuhler, Aleksey Tetenov, Michela Tincani, Harrison Wheeler and Danny Yagan for insightful conversations. This project also benefited from constructive comments from participants at numerous conferences and seminars. Special thanks are due to Mauricio Prado, Battista Severgnini, and Copenhagen Business School for their generous support in facilitating access to the data. All errors are my own. I gratefully acknowledge financial support from the GSEM - IEE and the Fonds Général at the University of Geneva.

1 Introduction

Parents shape children's lives. However, family influence extends beyond parental investments, affecting the social environment children are exposed to. The relevance of parental investments has been central to economic research since Becker and Tomes (1979) and substantial persistence in economic disparities across generations is documented in diverse contexts (Mazumder, 2005; Corak, 2013; Chetty et al., 2014). At the same time, families sort across neighborhoods and schools (Landersø and Heckman, 2017; Abdulkadiroglu et al., 2020; Eshaghnia et al., 2023). As a consequence, children from advantaged families are more likely to meet peers from similarly advantaged backgrounds (Chetty et al., 2022a,b). Whether and how this differential exposure reinforces the transmission of economic inequality is the central question of this paper.

I address three related questions. First, does peer exposure shape adult outcomes? Exploiting within-school, across-cohort variation in peer composition (Hoxby, 2000) for the universe of Danish high-school students, I find that a \$1 increase in schoolmates' average parental earnings raises annual adult earnings by \$0.10. Second, how much does peer exposure contribute to intergenerational persistence? The answer depends on how much peers matter and on how closely peer composition mirrors parental background. Peer influence is important, while Danish high schools are weakly segregated. As a result, exposure to schoolmates widens the earnings gap between children of top- and bottom-tercile parents by only 3.6%, while that contribution would be 11.2% under U.S. sorting rates. Third, why does peer exposure matter? Beyond shaping human capital, schoolmates become connections on the labor market. They facilitate connected hirings, and their career advancement spills over onto each other's wages. A dynamic mediation analysis shows that this channel accounts for 13% of the return to peer exposure.

These findings extend the notion that segregated interactions affect lifetime outcomes by shaping productivity (Loury, 1976; Benabou, 1993; Durlauf, 1996; Fogli et al., 2026), showing that social interactions affect *access* to opportunities too. As such, they suggest that policies increasing interactions across socioeconomic groups may improve intergenerational mobility. Moreover, the evidence on the role of job access via personal connections supports

the idea that social ties affect matching of workers to firms, with potential implications for overall match quality (Calvó-Armengol and Jackson, 2007; Bolte et al., 2026) that should be accounted for when considering desegregation policies.

The population of this study is the universe of Danish students enrolling in non-vocational high schools from 1988 to 2000. For each student peer quality is measured using the distribution of schoolmates' parental earnings and employment histories are observed through age 39. Linking these records makes it possible to observe family resources, school composition, and adult outcomes for the same individuals. The main outcome is average individual earnings between ages 35 and 39.

To study the effect of peer exposure, I adopt a linear-in-means model (Manski, 1993) exploiting school-specific, cohort-to-cohort fluctuations in peer composition (Hoxby, 2000). Identification faces two main concerns. First, students who would experience high earnings even absent peer exposure might select into better groups. Second, correlated shocks may affect both peer composition and future earnings. For example, improving local economic conditions could simultaneously attract higher-income families and improve students' later labor market opportunities.

To address these concerns, I exploit for identification within-school across cohorts fluctuations in school composition that are arguably unanticipated: students might sort into schools but are unlikely to forecast year-to-year marginal changes in the pool of enrolled schoolmates. Consistent with this argument, I show that residual fluctuations are uncorrelated with individual characteristics and show no serial correlation within schools. I also find that teacher quality does not respond to these fluctuations and that the estimates remain robust to controls for changing local economic conditions through municipality-by-cohort fixed effects. Together, these checks support interpreting residual changes in peer composition as a source of exogenous variation in exposure.

Peer exposure is consequential, in particular for children of higher-earnings parents. Each additional dollar in schoolmates' parental earnings raises adult earnings by \$0.13 for students in the top parental earnings tercile, compared with \$0.07 for those in the bottom tercile. Returns are also larger for males than females (\$0.12 versus \$0.09) and for students in schools below the median size of 60 students than those in larger schools (\$0.11 versus \$0.08).

Peer exposure reinforces intergenerational persistence the more children sort by parental background and the larger the effect of peer exposure. In Denmark, returns to exposure are large (as much as one third of the correlation between parental and children's earnings), while segregation across schools by parental earnings is limited. As a consequence, differential schoolmate exposure accounts for 3.6% of the difference in adult earnings between children from the top and the bottom tercile of parental earnings. Comparing the Danish sorting rate across schools and neighborhoods with estimates of residential sorting by U.S. Census tract obtained from publicly available data from the Opportunity Atlas (Chetty et al., 2026), I illustrate this magnitude. Had Denmark experienced the residential US sorting rate, peer exposure would have contributed to 11.2% of the same difference in adult earnings.

I then investigate why exposure matters. Consistent with a long-standing literature on the role of interactions for social mobility (Loury, 1976; Benabou, 1993; Durlauf, 1996; Fogli et al., 2026) and on the impact of peer effects in education (Epple and Romano, 2011), I find that peer exposure raises education achievement, increasing the probability of obtaining university education and shifting towards higher-returns fields. However, returns to employment outcomes are larger than predicted by improved education, with hourly wages increasing by 0.9% and firm pay premia (Abowd et al., 1999) by 15.4% per standard deviation in peer exposure. As these results are consistent with multiple channels, I develop two different designs to isolate the role of access on the labor market.

First, conditional on beginning a new job, school-based ties at a given firm increase the probability of being hired there by 68%. As observed job transitions confound network effects and shared characteristics between the mover and the peer acting as a tie at the destination firm, I adapt the almost-coworker design (Glitz and Vejlin, 2021; Harmon et al., 2025) to my setting to separate network effects. In practice, I compare hiring rates at firms employing *actual* schoolmates with hiring at firms employing only *almost* schoolmates: students from the same school in adjacent cohorts. Hence, I exploit the fact that almost schoolmates (ties that would have been realized had the student enrolled one year earlier or later) offer an informative control group sharing alternative sorting factors, such as geographic proximity between schools and firms or the match between skills taught at school and those demanded by firms, but are less likely to be a social contact.

Second, former schoolmates' career advancements generate wage spillovers. Exploiting the timing of peers' individual promotions to managerial positions in a staggered difference-in-differences model that compares wage growth among peers of a worker promoted in a given year with that among peers of a not-yet-promoted manager (Callaway and Sant'Anna, 2021), I estimate that a peer's promotion increases average wages by up to 2% in the next 8 years. This finding is consistent with peers providing outside options (Caldwell and Harmon, 2019; Caldwell and Danieli, 2024) that support wage growth through on-the-job search (Postel-Vinay and Robin, 2002; Bagger et al., 2014).

Finally, I show that earlier peer promotions explain 13% of the returns to exposure. To do so I extend recent advancements from the literature on mediation analysis (Schenk, 2025) to a case in which the mediator (peer promotion) has a staggered implementation, nesting it into the peer effect design of this paper. Within-school, across-cohort variation identifies the effects of exposure on both earnings and the timing of peer promotion, while an adjusted difference-in-differences design estimates peers' promotion effects conditional on exposure. Identification builds on assumptions similar to those underlying the previous results: exogeneity of school-composition fluctuations and parallel trends between individuals whose peers are promoted one year apart. An additional portability assumption links the two designs: shifts in peer promotion timing must have the same earnings effects whether induced by changes in peers' parental background or by other reasons.

This paper contributes to the existing literature along three different lines. First, it contributes to a large literature investigating the extent and drivers of intergenerational persistence. Theoretical work highlights how socioeconomic segregation can perpetuate inequality by separating families into social environments with different opportunities (Loury, 1976; Benabou, 1993; Durlauf, 1996). Related evidence documents geographical variation in mobility (Chetty et al., 2014) and higher upward mobility in US communities with more friendships across socioeconomic groups (Chetty et al., 2022a,b). I contribute by providing causal evidence on the role of peers' parental backgrounds in adult earnings and quantifying its contribution to persistence.

Second, I show that high-school exposure shapes access to jobs through social ties, complementing research on peer effects on learning, aspirations, risky behavior, and educational

choices. Two strands of research motivate this channel. A growing literature shows that access to employers and high-wage jobs is transmitted from parents to children (Kramarz and Skans, 2014; Staiger, 2026; Ventura, 2024; San, 2026; Dobbin and Zohar, 2026; Forsberg et al., 2026). Another shows that ties formed among coworkers (Hensvik and Skans, 2016; Glitz and Vejlin, 2021; Saygin et al., 2021) and college peers (Zimmerman, 2019; Michelman et al., 2022; Zhu, 2025; Ost et al., 2026) facilitate hiring. I link the two, additionally showing that high-school peers are also relevant social ties. Because high-SES students have privileged access to high-wage jobs, ties with them become a channel of job access in their own right: opportunities are transmitted not only vertically, from parents to children, but also horizontally, across peers. Relative to existing evidence on schoolmates (Eliason et al., 2023), which compares connected and unconnected workers moving between the same pair of firms, I adapt the almost-coworker design (Glitz and Vejlin, 2021; Harmon et al., 2025) to compare potential destinations for the same worker, and find similar effects under different identifying assumptions. Closely related work by Fischer et al. (2026) exploits random assignment to tutorial groups at Copenhagen Business School and finds consistent results on job access among alumni. I complement their analysis by studying ties formed earlier, in high school, across a broader population, by showing that these ties remain valuable throughout the career, and by quantifying how much improved labor market connections mediate the long-run earnings returns to peer exposure.

Third, I design and apply a novel mediation technique to study the mechanisms through which peer exposure affects outcomes over time. The framework allows baseline exposure to affect outcomes both directly and indirectly through a mediator whose timing varies conditional on exposure. It builds on the recent developments in the causal mediation literature (Imai et al., 2010; Huber and Frölich, 2020), with the work from Schenk (2025) being the main starting point. I adapt this framework to a setting in which within-school, across-cohort variation provides conditionally exogenous peer exposure, while the mediator occurs at different times across groups. Drawing on and extending the staggered difference-in-differences approach of Callaway and Sant'Anna (2021), I combine these sources of variation to quantify how exposure-induced shifts in peer promotion timing translate into subsequent earnings gains.

The rest of the paper is organized as follows. Section 2 describes the data and the sample construction. Section 3 presents the empirical strategy and the main results on peer effects. Section 4 discusses the implications for intergenerational mobility. Section 5 investigates the role of networks in shaping access to higher-paying jobs. Section 6 presents a formal mediation analysis to quantify the contribution of networks to peer effects. Section 7 discusses the results and Section 8 concludes.

2 Institutional Setting and Population Characteristics

2.1 Danish High Schools

Danish students complete compulsory education by the 9th grade, typically at age 16. Attendance of an extra 10th grade is optional. After finishing lower secondary education, students can choose to either enroll in a high school that grants access to university, attend a vocational school, or discontinue their education. Non-vocational high schools are the setting of this paper.[1]

High-school programs range from 2 to 3 years, depending on the track chosen, and aim to prepare students for tertiary education or entry into the labor market. They are organized into four principal tracks, each featuring distinct curricula. These tracks are tailored to prepare students for university-level studies or provide more technical and business-oriented training. Each track includes compulsory courses and elective subjects, allowing for some degree of curricular flexibility.

Admission to high school is conditional upon the successful completion of lower secondary education. Students submit ranked preferences for high schools and tracks. School placement is determined at the national level based on students' residential addresses: when preferred schools are oversubscribed, students are assigned to a similar school within their district. Although high schools are self-governing institutions, their funding primarily comes from state transfers, and tuition fees are either absent or minimal.[2]

2.2 Sample Selection

Administrative registers covering the universe of the Danish population from 1980 to 2024 are the primary data sources of this paper. The sample includes the students who enrolled in a Danish high school from 1988 to 2000. Parents and schoolmates are identified through family and school records, respectively. Additionally, earnings are tracked annually using tax records, while labor market outcomes are observed each year through employer-employee

[1] 45.6% of the children born in the same years as those included in the sample have enrolled in a non-vocational high school by the age of 19.

[2] According to OECD statistics, 3% of the expenditure on upper and lower secondary education is directly financed by households (Nusche et al., 2016).

matched data for both children and their parents.

Of the 465,430 students who enrolled in Danish high schools between 1988 and 2000, I exclude 34,511 (7.4%) who enrolled before age 15 or after age 18, and 2,019 (0.4%) from cohorts in which more than 50% of students enrolled outside this age range. Among the remaining students, parental earnings are missing for 1,203 students and adult outcomes are not observed for 11,646 individuals. Finally, I exclude 53 students with no eligible schoolmates after these restrictions and 5,018 who attended schools with fewer than four consecutive years of observation. The final sample consists of 410,980 students from 447 schools across 13 cohorts.

2.3 Measures of Earnings and Peer Exposure

Individual Earnings Earnings are measured as the average annual earnings from the main occupation and self-employment before tax from age 35 to 39, as reported in tax registers. For ease of interpretation, earnings are deflated for inflation and reported in 2015 USD.

Individual Earnings - Ranks To address concerns due to differences in lifecycle wage profiles, I construct percentile ranks of individual earnings relative to the distribution of children born in the same year.[3]

Parental Earnings For each parent, I measure earnings as the average yearly earnings from employment and self-employment when the child is 10 to 16 years old. I then construct parental earnings as the average among the parents of each child.

Parental Earnings - Ranks I construct percentile ranks of parental earnings relative to the distribution of children born in the same cohort.

Peer Group Schoolmates are identified as the children who enrolled at the same school and track in the same cohort as the child of interest, excluding the child herself.

[3]Evidence from Sweden suggests that the rank-rank correlation is less sensitive to measurement issues due to different lifecycle wage profiles (Nybom and Stuhler, 2016).

2.4 Descriptive Statistics

The main characteristics of the sample are reported in Table 1. The sample includes $410,980$ students enrolled in 13 cohorts across 447 schools. The average student has 98 schoolmates; her parents earn $\sim \$48,500$ per year, and she earns $\sim \$60,000$ per year by the age of $35-39$. Comparing children from different backgrounds, two main facts emerge from the table. First, children from higher-income parents tend to have higher earnings themselves. The difference in earnings between children from the top and the bottom tercile is substantial: $\$13,747.54$ or 11.44 percentile ranks. Second, differences in parental earnings are mirrored in schoolmates' parental earnings. The difference between the schoolmates' parental earnings of children from the top and the bottom tercile is $\$4,639.28$ and 5.96 percentile ranks, amounting to $\sim 10\%$ of the difference in their own parental background.

[Table 1 HERE]

Finally, Figure 1 plots the conditional means of schoolmates' parental earnings (blue dots) and neighbors' parental earnings (orange circles), by percentile of own parental earnings.[4] Differences in own parental earnings are reflected in schoolmates' family background mostly at the top of the distribution, and the same pattern is observed for neighbors, a measure of residential sorting. These facts are informative on the institutional context of Danish high schools: free access to education is likely to mitigate segregation, especially at the bottom of the earnings distribution, while residential segregation potentially drives sorting into more homogeneous peer groups at the top of the distribution.

[Figure 1 HERE]

[4]For each student, I define as her neighbors the individuals who are born in the same year as the focal individual and live in the same municipality at the age of 16.

3 Peer Exposure

3.1 Main Specification

The main goal of this section is to estimate the effect of exposure to schoolmates on adult earnings. Formally, we are interested in the causal relation between schoolmates' parental earnings ($\{X_j\}_{j=1}^{N_i}$, where $j \neq i$ indexes the schoolmates of individual i and N_i is their number) and individual earnings (Y_i). Two choices determine the research design: the functional form governing the relation between peers' parental earnings and individual earnings, and the source of variation in peers' parental earnings used for identification. I discuss these two choices in turn.

I use a linear-in-means model (Manski, 1993) as the baseline specification. I therefore construct a leave one out measure of schoolmates' parental earnings, ($\bar{X}_{-i} = \frac{1}{N_i} \sum_{j=1}^{N_i} X_j$ with $j \neq i$), and estimate the linear response of individual earnings to this measure. Some implications guide the interpretation of the results. First, the baseline model might not capture the full complexity of peer effects, aggregating heterogeneous or nonlinear effects. I investigate those separately. Second, the model identifies the effect of exposure, not interactions which are not observed in the data.[5] Third, the model identifies the exposure effect as a combination of the direct effect of peers' parental background and the indirect effect emerging from peers' achievements (*exogenous* and *endogenous* peer effects in Manski (1993)). I argue that this is an informative parameter, as changing exposure would change both. Still, to shed light on mechanisms, I separate the role of peers' career success in section 5.

Random assignment of children to schools would serve as the ideal experiment to estimate exposure effects. In the absence of such an experiment, unobserved determinants of earnings might be correlated within groups because of endogenous sorting or group-level correlated shocks. For example, individuals sharing similar unobserved characteristics (such as ability) might sort in the same group or common shocks (such as changes in local economic conditions) might simultaneously affect group members and generate correlation in outcomes

[5]Evidence from US Facebook friendship networks shows that high-school exposure affects social connections: Chetty et al. (2022b) find that a 10 percentage point increase in the share of schoolmates from top-quintile parental socioeconomic backgrounds raises the share of such students among bottom-quintile students' within-cohort friends by 8.9 percentage points.

even in the absence of peer effects. To address these concerns, I exploit for identification year-to-year changes in school composition that are likely unanticipated by agents (Hoxby, 2000).[6]

To do so, I estimate the following model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 \bar{X}_{-i} + W_i'\delta + \kappa_{s(i)} + \lambda_{s(i)} c(i) + \varepsilon_i. \tag{1}$$

Here, $c(i)$ and $s(i)$ denote the enrollment cohort and the school of individual i, respectively. Hence, $\kappa_{s(i)}$ is a set of school fixed effects, and $\lambda_{s(i)}c(i)$ represents school-specific time trends. Finally, W_i is a vector of predetermined individual characteristics which includes indicator variables for gender, year of birth and cohort of high-school enrollment. The main coefficient of interest (β_2) captures the marginal effect of average peer's parental earnings on individual earnings.

The identifying assumption is that, conditional on school-specific time trends and included controls, residual variation in schoolmates' parental earnings is not correlated with unobservable determinants of earnings (ε_i). Accounting for school-specific time trends restricts the identifying variation to year-to-year changes in school composition. As a consequence, the assumption is compatible with a model in which students are allowed to choose schools based on their (expected) peer quality, but are unable to forecast marginal deviations that emerge from finite-sample fluctuations around each school trend. Moreover, the high year-to-year volatility of these residual shocks makes them unlikely to be related to underlying shifts in school inputs or local economic conditions. In the following section, I document this variation and provide evidence supporting the identifying assumption.

3.2 Fluctuations in School Composition

Table 2 describes the variation in school composition. The first four columns report the average characteristics of schoolmates by tercile of own parental earnings and in the full sample. The last three columns report the standard deviation of each measure once residu-

[6]This design, introduced by Hoxby (2000), has been thereafter adopted to study peer effects in different domains: among schoolmates in education, by Black et al. (2013); Carrell et al. (2018); Brenøe and Zölitz (2020); Cattan et al. (2025); Meluzzi (2024), or among coworkers, by Cornelissen et al. (2017).

alized on progressively richer sets of fixed effects. In the baseline specification, which includes school fixed effects, cohort fixed effects and school-specific time trends, the remaining variation is approximately one-third of the total variation in schoolmates' parental earnings. This remaining variation is economically meaningful. A one-standard-deviation change in the residual share of peers from a given tercile corresponds to about 5 percentage points, roughly 15% of the average share of 33% (or 5 students in the average group size). This is the identifying variation used in the main specification.

[Table 2 HERE]

Table 3 shows the volatility of year-to-year fluctuations in school composition. For each school, I estimate the first-order autocorrelation of schoolmates' parental earnings across consecutive cohorts and report the share of schools in which it is statistically different from zero. Raw composition is persistent: for 17% of the schools in the sample the hypothesis of no serial correlation is rejected at the 10% significance level, above what is expected by chance. Including additively school and cohort fixed effects greatly reduces the share of schools with serial correlation, but their rates are still (slightly) above the rates expected by chance (10.27% vs 10%, 6.08% vs 5% and 1.68% vs 1%). This is consistent with some schools following potentially predictable secular trends in their composition. Once school-specific time trends are included, rejection rates fall well below their distribution under the null hypothesis of no correlation (9.4%, 3.8% and 0.4%). These fluctuations are therefore statistically unpredictable and transitory, making them unlikely to drive enrollment decisions or to be correlated with shifting local economic conditions.

[Table 3 HERE]

Figure 2 concludes this section by showing that fluctuations in school composition are uncorrelated with students' predetermined characteristics. The logic resembles a balance test: if residual variation in schoolmates' parental earnings is as good as randomly assigned, it should not predict the characteristics of the students exposed to it. Figure 2 reports coefficients from separate regressions of schoolmates' parental earnings on predetermined predictors of individual earnings, before (orange circles, left axis) and after (blue diamonds, right axis) residualizing on school fixed effects, cohort fixed effects and school-specific time

trends.[7] Consistent with the sorting of families across schools (Landersø and Heckman, 2017), the raw association between own and schoolmates' parental earnings is positive and large (0.239, s.e. 0.025). Once school composition is measured as a deviation from school-specific trends, the same coefficient falls to −0.001 and is not statistically different from zero. All other characteristics follow the same pattern: raw coefficients range from 0.227 (father's college degree) to −0.112 (father's high-school diploma), while residual coefficients never exceed 0.003 in absolute value, two orders of magnitude smaller.[8] Students with different observable characteristics therefore do not sort into cohorts with different deviations from expected composition. Appendix Figure A.1 repeats the exercise for school inputs: teachers' wage, age, work experience and the share of new hires are unrelated to residual school composition, suggesting that schools do not adjust to year-to-year, transitory, fluctuations in their composition.

[Figure 2 HERE]

[7] All regressions control for the school-level average of parental earnings. This follows the test developed by Guryan et al. (2009), which corrects for the mechanical negative correlation between an individual's characteristics and the leave-one-out average of the same characteristic among their schoolmates. Table 4 reports the same estimates in table format, together with an intermediate specification including only school and cohort fixed effects.

[8] Due to the precision of the estimates, a few of them are statistically significant, but their size is negligible. Including all these characteristics as controls in the main specification leaves the coefficient of interest unchanged (Appendix Table A.2).

3.3 Main Results

Table 5 displays the main results of the paper, reporting OLS estimates of model (1). Earnings are measured in 2015 USD in columns (1)–(4) and as percentile ranks in columns (5)–(8).

Columns (1) and (5) document intergenerational persistence without accounting for peer exposure. On average, children whose parents earn $\$1,000$ more earn \$249 more as adults. In ranks, the rank–rank slope (Chetty et al., 2014) is 0.16: a 10-percentile increase in parental earnings is associated with a 1.6-percentile increase in children's adult earnings.[9]

Schoolmates' parental earnings are at least as strongly associated with adult earnings as own parental earnings. Comparing children with the same parental earnings, those exposed to schoolmates from higher-earning families earn substantially more (columns 2 and 6); in levels, the coefficient on schoolmates' parental earnings (0.307) even exceeds the coefficient on own parental earnings (0.230). However, this association captures both the causal effect of exposure and endogenous sorting of higher achieving students in higher quality peer groups.

When only idiosyncratic deviations from expected school composition are exploited for identification in columns $(3, 4)$ and $(7, 8)$, the main result of the paper emerges: peer exposure has a strong, statistically significant causal impact on lifetime earnings. Columns (3) and (7) include school and cohort fixed effects, the standard specification in the literature, while columns (4) and (8) account for school specific time trends, and are the preferred specification. Under this richer specification, an increase of \$1000 in peers parental earnings generates an increase of \$96 in adult earnings, making peers' parental background as effective as 38% of parental background.[10]

[Table 5 HERE]

[9] For comparison, Chetty et al. (2014) estimate an income rank–rank slope of 0.34 for the United States. For Denmark, Landersø and Heckman (2017) report a wage-earnings rank–rank slope of 0.20 for the full population of children born in 1973–1975, with parental earnings averaged over the child's ages 7–15 and children's earnings measured over three years at ages 35–39. My estimate differs in two respects. First, parental earnings are measured over a shorter window (the child's ages 10–16). Second, my sample includes only students who enrolled in high school.

[10] This estimate is conservative: without school trends the effect is \$123, and the residual imbalances in that specification are economically negligible (Table 4).

3.4 Differences in Returns to Exposure

Do different students benefit differently from changes in average schoolmates parental earnings? Table 6 reports estimates of a modified version of model (1) in which exposure is interacted with student characteristics. Results show that own parental background is consequential. Each additional dollar in schoolmates' parental earnings raises adult earnings by \$0.13 for students in the top parental earnings tercile, compared with \$0.07 for those in the bottom tercile, although the difference is imprecisely estimated. A similar pattern is found when students are split by parental education: those with at least one parent with a university degree gain \$0.13 vs \$0.04 for the others. Overall, more advantaged students benefit to a larger extent from similarly advantaged peers.

Returns are also larger for males than females (\$0.12 versus \$0.09) and for students in schools below the median size of 60 students than those in larger schools (\$0.11 versus \$0.08). The larger impact in smaller groups, although not statistically significant, is consistent with exposure being a stronger shifter of social interactions in smaller groups, where fewer potential contacts limit the scope for students to sort into homogeneous friendships. Finally, the result is not specific to urban environments: the effect is 0.11 for schools in Copenhagen, Aarhus and Odense (the three largest Danish cities) and 0.09 elsewhere.

[Table 6 HERE]

3.5 Nonlinearity and Functional Form

Does the leave-one-out average of schoolmates' parental earnings adequately summarize the treatment of exposing a student to a different distribution of peers? The linear-in-means model (1) restricts the effect of a change in peer composition to be independent of its baseline level and symmetric to changes at the top and the bottom of the peer distribution.[11] I test these two implications on my sample.

First, with linearity the effect of exposure should not depend on the average composition of the school. Figure 3 reports estimates of eq. (1) in which the effect is allowed to vary by

[11] A large literature documents departures from this benchmark (Feld and Zölitz, 2017; Carrell et al., 2018) and proposes alternative microfoundations (Brock and Durlauf, 2001; Boucher et al., 2024; Tincani, 2026).

school baseline earnings, effectively splitting schools by tercile of students' parental earnings averaged across all cohorts. Point estimates are somewhat larger in higher-earnings schools, but the differences are small and not statistically significant ($p = 0.86$).[12]

Second, if the exposure effect is linear, the effect of replacing peers from the middle of the distribution with poorer or richer peers should be symmetric and proportional to the difference in their parental earnings. In Figure 4, I report estimates from a modified version of model (1) after replacing the leave-one-out mean with the shares of peers whose parental earnings fall in the bottom and top terciles of the distribution. With the middle tercile omitted, the coefficients capture the effects of replacing middle-tercile peers with bottom- or top-tercile peers, respectively. The dashed line reports the effect implied by the linear model. The estimated effects of both substitutions lie close to this prediction in the full sample and for each group of students.[13] The share estimates are imprecise, so moderate nonlinearities cannot be ruled out. However, failing to reject the main implications of a linear model, I retain it as a parsimonious and empirically adequate description of how peer composition affects earnings.

[Figure 3 HERE]

[Figure 4 HERE]

Overall, three main results emerge from this section. First, peer parental background is consequential: each extra dollar of schoolmates' parental earnings increases adult earnings by \$0.10. Second, this effect is comparable to parental influences, with its point estimate being as large as one third of the correlation between children and parental earnings. Third, while everyone benefits from improved peer exposure, influence is concentrated at the top of the parental earnings distribution.

[12] It is important to notice that this does not rule out potential nonlinearities as the comparison is performed across schools. Nonlinear effects might be offset by differences in returns to exposure by school due to other factors correlated with baseline school type, which are not separately identified in this design.

[13] For students from the top tercile, the cost of replacing middle-tercile peers with bottom-tercile peers is larger than predicted, and the gain from top-tercile peers smaller, but the difference between the two marginal effects is not statistically significant.

4 Peer Exposure and Intergenerational Mobility

Children from higher earnings parents earn more as adults and grow up next to different peers in their youth. How much of the first difference is due to the second? Two forces determine the answer: how much peers matter, and how closely peer composition mirrors parental background. The previous section illustrates the effects of peer exposure in a specific setting: high-school peers. This section combines the peer effects estimated above with observed sorting across schools to quantify the contribution of schoolmate exposure to intergenerational earnings persistence.

4.1 A Linear Decomposition Framework

Consider the rank-rank correlation in earnings (Chetty et al., 2014) between parents and children. This is the slope α in $Y_i = \alpha_0 + \alpha X_i + u_i$, estimated in column (5) of Table 5. This parameter measures *relative* persistence, accounting for how much differences in parental earnings are mirrored in their children. Suppose that childrens' earnings also depend on schoolmates' parental earnings $\bar{X}_{-i}$, as in eq. 1, and that peers' parental earnings are linearly related to one's own, as in $\bar{X}_{-i} = \rho_0 + \rho X_i + v_i$. The usual omitted-variable formula links intergenerational persistence to the effect of peer exposure as follows:

$$\alpha = \beta_1 + \beta_2 \ \rho + L, \tag{2}$$

where L is the projection of the controls included in eq. (1) on parental earnings.[14] The rank–rank slope thus combines three channels: the parental gradient conditional on peers and schools (β_1), the causal return to exposure combined to the extent of sorting ($\beta_2 \ \rho$) and the parental gradient on access to schools and other controls included in the model (L). The share of persistence due to peer exposure is thus $\frac{\beta_2}{\alpha} \times \rho$ and two forces drive it: the ratio of the (causal) effect of peer exposure to parental gradient in earnings ($\frac{\beta_2}{\alpha}$) and the realized sorting (ρ). The estimates from Table 5 show that the first component is large: peers are

[14] Let $L_i = W_i'\delta + \kappa_{s(i)} + \lambda_{s(i)}c(i)$ collect the controls and the school-specific intercepts and trends, and $L = Cov(L_i, X_i)/Var(X_i)$. The residual ε_i is orthogonal to X_i by construction, so eq. (2) is an accounting identity rather than an additional assumption.

about 29% as influential as parents ($\hat{\beta}_2/\hat{\alpha} = 0.29$) when earnings are expressed in percentile ranks. Unlike most settings, the full population of schoolmates is observed here, so ρ can be estimated directly regressing peers' parental earnings on own earnings: in Danish high schools, $\rho = 0.09$ (Table 7). Combining the two estimates, exposure to schoolmates accounts for about $0.29 \times 0.09 \times 100 = 2.5\%$ of intergenerational persistence among Danish high school students.

The small share combines a sizable return to peer exposure with very low average segregation across Danish high schools. This raises two questions: how much would peers contribute in a more segregated setting and how much does the linear approximation miss? I address them next.

4.2 Benchmarking against U.S. Sorting

Average sorting across schools is low in Denmark. How does it compare with other settings? Direct estimates of ρ are rarely available, because they require data linking individuals to their peers and to their peers' parental earnings. However, ρ is closely related to the share of the total variance in parental earnings that lies between groups. Intuitively, if children share similar parental backgrounds within each group, most of the total variation is explained by differences across group averages. An estimate $\hat{\rho}$ therefore can be constructed as the ratio of the variance of group means over the population variance of parental earnings. Appendix B derives this relation and shows that ρ necessarily lies in $[\hat{\rho} - b,\ \hat{\rho}]$, where the bound b accounts for the fact that $\bar{X}_{-i}$ excludes the child herself and is decreasing in group size.

Table 7 assesses the accuracy of this approximation in Denmark, where both ρ and $\hat{\rho}$ can be estimated directly for schoolmates and for neighbors living in the same municipality, a measure of residential sorting. The direct and variance-based estimates are close, and ρ lies within the bounds in both cases. Applying the same approach to publicly available Opportunity Atlas data (Chetty et al., 2026) on parental earnings across U.S. census tracts yields $\hat{\rho}^{US} = 0.282$, with bounds $[0.277, 0.282]$. Sorting across Danish high schools is thus relatively limited, roughly one-third as large as residential sorting across U.S. census tracts.

With this estimate in hand, a different counterfactual can be assessed. If Danish school sorting matched the U.S. residential sorting rate, holding the Danish return to peer exposure

fixed, the contribution of peer exposure to intergenerational persistence would be $0.29 \times 0.28 \times 100 = 8.1\%$.

[Table 7 HERE]

4.3 Nonlinear Sorting and Heterogeneous Returns

The linear decomposition averages over two dimensions of heterogeneity. Danish sorting is concentrated at the top, and the return to exposure increases with parental background. I conclude by accounting for those two factors in the quantification of the role of exposre in intergenerational mobility.

[Figure 5 HERE]

Panel (a) of Figure 5 plots peers' parental earnings against own parental earnings under three scenarios. The convex profile is the observed sorting in Denmark: children from the bottom and the middle of the parental distribution are exposed to similar peers, and segregation mainly emerges at the top tercile. The linearly increasing profile represents the U.S. degree of sorting estimated in Section 4.2. And finally, the flat line represents the benchmark of no sorting, in which every child is exposed to the same average peer quality. Panel (b) reports the effect of sorting on adult earnings at each percentile of the parental distribution, estimated by multiplying the percentile-specific[15] effect of peer exposure by the difference between the child's peer exposure and the no-sorting benchmark in panel (a).

Again, U.S. sorting rates imply the largest effects: about +\$1,550 at the top of the parental distribution and −\$800 at the bottom. The gains at the top exceed the losses at the bottom even though the U.S. sorting benchmark is linear, because the return to exposure increases with parental earnings. Danish sorting implies much smaller effects that are concentrated at the top: −\$120 at the bottom and +\$1,050 at the very top, as a consequence of the convex sorting profile corresponding to increasing returns to exposure by parental background. Panel (c) translates these effects into the earnings gap between children of top-

[15]The return to exposure is smoothed by estimating a version of model (1) where the effect of peers' parental earnings is allowed to vary quadratically in parental earnings rank, as reported in Figure 6. Using the non parametric estimates of tercile-specific effects in Table 6 leaves the results unchanged.

and bottom-tercile parents. Without sorting, the gap would be \$13,240. Observed Danish sorting raises it to \$13,715 (+3.6%), and U.S. sorting rate to \$14,726 (+11.2%).

Three considerations qualify these magnitudes. First, they concern high-school peers, one component of children's broader social environments. Second, they capture the effects of exposure rather than actual interactions, leaving open how changes in cohort composition translate into friendships and other social ties. Third, this is a partial-equilibrium accounting exercise: it holds parental investment, school inputs, and other determinants of earnings fixed, deliberately abstracting from responses and changes in inputs that a policy-induced redistribution of students might generate. These considerations are explored further in the discussion section. Under these qualifications, peer exposure accounts for 2.5% to 3.6% of observed intergenerational earnings persistence in Denmark. Holding the Danish return to exposure fixed, this contribution would rise to about 8% of observed Danish persistence under U.S. levels of sorting. Peer exposure therefore contributes substantively to the transmission of inequalities from one generation to the next, and increasingly so with the extent to which parental differences are mirrored in peers' socioeconomic background.

5 Peers: Connections on the Labor Market

Exposure to higher-SES schoolmates raises adult earnings, but the same reduced-form effect can arise from different channels with different implications. One possibility is that peers shape learning. Sitting in class with more advantaged schoolmates may improve learning (Fruehwirth and Gagete-Miranda, 2019), induce effort (Tincani, 2026; Bursztyn et al., 2018), or influence teachers' instruction (Salvati, 2023). In these scenarios, peer effects are part of the learning process itself and raise students' productivity as adults. Another possibility is that exposure shapes networks. Higher-SES schoolmates, who are more likely to work in high-wage jobs (Staiger, 2026; Forsberg et al., 2026; San, 2026), might remain social ties on the labor market, facilitating access to higher-paying jobs via referrals (Montgomery, 1994; Eliason et al., 2023; Bolte et al., 2026). Importantly, this second channel does not necessarily involve changes in productivity.

Learning and connections are not mutually exclusive, and both are likely at work: the same peers who facilitate learning today may become valuable social ties tomorrow. Before formally isolating the network channel in Section 6, I establish three facts here. First, peer exposure improves both educational attainment and labor market outcomes, but the educational gains are too small to explain the full earnings returns. Second, when individuals switch jobs, connections to former schoolmates open doors to their workplaces. Third, career dynamics are connected: wages rise as former schoolmates provide improved outside options after a promotion.

5.1 Education and Job Access

Table 9 reports the effect of exposure on labor market and educational outcomes at ages 35–39. A one-standard-deviation increase in schoolmates' parental earnings raises annual earnings by \$779 (1.3%) and hourly wages by \$0.31 (0.89%). Exposure raises access to high wage firms: the pay premium of the firm where students work increases by 15.4%, with its largest impact at the top of the firm distribution: employment at top decile pay premia firms increases by 3.41%. Effects on education are also positive and significant, but their magnitude is more limited. Exposure raises the probability of completing high school

by 0.2 percentage points (0.23%) and of obtaining a university degree by 0.4 percentage points (0.6%). As academic high schools are designed to prepare students for university, the relevant margin may instead be the type of education students choose. To capture this intensive margin, I assign each student the average hourly wage that earlier cohorts with the same level and field of education earned.[16] This predicted wage rises by \$0.125 per standard deviation of exposure, compared with \$0.31 for actual wages. Even under a broad definition of education, then, changes in attainment account for at most about 40% of the wage effect.

[Table 9 HERE]

Importantly, these estimates measure effects on potential mediators, and they do not by themselves separate mechanisms. Consider university enrollment. If exposure to advantaged peers leads students to enrol in more selective programmes, the earnings gain could reflect what they learn there, the networks they form there, or both. Once again, the same (causal) evidence is consistent with very different underlying mechanisms. For this reason, the remainder of this section moves from outcome-by-outcome comparisons to designs that isolate the network channel directly.

5.2 Connected Hires

Social ties are consequential for job access. Employees with limited knowledge about applicants might exploit incumbent employees' information and job seekers might obtain information about the job from social contacts (Montgomery, 1994; Calvó-Armengol and Jackson, 2007; Dustmann et al., 2015; Hensvik and Skans, 2016; Harmon et al., 2025). Is this also true for connections formed in high school?

The main identification concern is that social connections are not randomly assigned. In our context, former schoolmates may share characteristics that influence their match with a given firm, such as skills that make them attractive applicants or a geographical location that makes the firm an attractive destination. Observed hiring at connected firms therefore combines the influence of social ties with that of shared characteristics.

[16] Education is classified using Statistics Denmark's mapping of completed educations into ISCED-97. Level is the ISCED level of the education, and field is the first digit of the ISCED field of education, which distinguishes nine broad fields: general programmes; education; humanities and arts; social sciences, business and law; science; engineering, manufacturing and construction; agriculture; health and welfare; and services.

To circumvent this identification issue, I compare hiring probabilities across different destinations for the same student. I classify as connected those firms employing an actual schoolmate and compare the probability they hire the focal student with a closely comparable set of firms who employ potential ties who are similar to the real ones, but less likely to be formed: *almost schoolmates* who enrolled in the same high school in the year before or after the focal student. The relevance of the contrast relies on the observation that social ties are more likely to be formed within each cohort and its validity on the assumption that determinants of sorting to each specific firm are on average the same between *actual* and *almost schoolmates*. This adapts designs used to study coworker networks (Glitz and Vejlin, 2021; Harmon et al., 2025), where almost coworkers are individuals who joined the focal worker's firm shortly after her departure or left shortly before her arrival.

Empirically, I aim to measure whether a worker is more likely to begin a new job at a *connected* firm,[17] where an actual schoolmate is already employed, than at a firm employing only almost schoolmates. To do so, I estimate the following dyadic regression among workers beginning a new job:

$$H_{if} = \alpha_i + \delta_1 C_{if} + \delta_2 AC_{if} + u_{if}, \tag{3}$$

where each observation is an individual–firm pair. The outcome H_{if} indicates whether individual i is hired by firm f. The indicator C_{if} equals one if the firm employs an actual schoolmate of i, while AC_{if} equals one if it employs an actual or almost schoolmate. These indicators distinguish three types of pairs: connected, almost connected only, and unconnected. Because AC_{if} includes actual connections, δ_1 measures the difference in predicted hiring probability between connected firms and firms with only almost connections. Finally, individual fixed effects α_i ensure that the comparison is made across potential destinations for the same worker.[18]

[17] Throughout this section, a firm is an establishment (plant), identified by the establishment identifier assigned by Statistics Denmark.

[18] Including individual-specific fixed effects de-means the model at the worker level, ensuring differences in network sizes are not driving the estimates, with each match probability being implicitly compared to a random matching benchmark. Appendix C describes the estimation procedure, which recovers the relevant OLS coefficient exploiting sparsity from properly adjusted group-firm counts, without loading the full set of dyads.

[Table 10 HERE]

Table 10 reports the estimates. Pooling hires between ages 25 and 39, a worker is 68% more likely to be hired by a firm where an actual schoolmate works than by a firm employing only almost schoolmates.[19] The comparison with almost connections matters: firms employing almost schoolmates are themselves about two orders of magnitude more likely to hire the worker than unconnected firms, reflecting the shared characteristics the design nets out.

Connections also open access to high-paying firms. A connection raises the hiring probability by 38% for students from high-earning families and by 44% for low parental earnings students at firms in the top tercile of pay premia. Connections partly follow homophily by parental background: a high parental background peer raises the hiring probability of a similar mover by 0.031 p.p. (41%) and of a low parental background mover by 0.023 p.p. (36%). Ties nonetheless are valuable across backgrounds. Rich peers significantly raise hiring for low-parental-earning movers, and poor peers do so for students from high-parental-earning movers (0.031 p.p., 54%). Finally, Figure 7 shows that school connections matter throughout the career: the increase is stable at 60–80% between ages 25 and 39.

[Figure 7 HERE]

Overall, these estimates show that social ties developed in high school are consequential for job search prospects. Former schoolmates open doors offering valuable opportunities to improve career outcomes.

5.3 Connected Opportunities and Outside Options

How important are the employment opportunities facilitated by social connections to sustain wage growth and career advancement? Those might be relevant for at least two reasons. First, transitioning to a new job exploiting a social tie might result in a wage increase if the new match is more favorable to the worker than the previous one. Second, alternative employment opportunities from social contacts ultimately shape workers' outside options

[19] I estimate the model both with individual-specific intercepts α_i and with a single intercept α. As in Harmon et al. (2025), the estimates of δ_1 and δ_2 do not change. I therefore report the single-intercept estimates and use $\hat{\alpha}$ to measure the baseline hiring probability at almost-connected firms, $\hat{\alpha} + \hat{\delta}_2$. Estimates with individual-specific intercepts are available upon request.

affecting the workers' bargaining position with her current employer, who has to raise the wage when a credible outside offer reaches an incumbent worker or lose the worker to the competing firm (Postel-Vinay and Robin, 2002; Bagger et al., 2014; Caldwell and Harmon, 2019).

Estimating the return to the outside options provided by former schoolmates is complicated by the fact that the schoolmates who may raise a worker's productivity through spillovers in human capital formation in school are the same peers who later act as social ties on the labor market. To solve this issue, I study career dynamics around sudden improvements in schoolmates' labor-market position: managerial promotions.

In a staggered difference in difference design (Callaway and Sant'Anna, 2021), I compare wage growth around each promotion event between peers of the promoted manager and control peers of not-yet-promoted managers.[20] Under the assumption that the average wage in the two groups would have evolved on parallel trends absent the promotion, this recovers the spillover of an individual promotion on former schoolmates' careers. To ensure that correlated timing of career advancement between those who get promoted and their peers does not drive the estimated effect, I restrict the comparison to individuals of the same age and flexibly control for their cohort, school type and school geographical location.

To do so, I estimate the following model of the evolution of wages by age:

$$W_{sc,a} = \alpha_{sc} + \alpha_a + \sum_{\ell \neq -1} \delta_\ell \mathbb{1}\left\{a - G_{sc} = \ell\right\} + \psi_{r(s),c,a} + \xi_{k(s),a} + \epsilon_{sc,a}, \tag{4}$$

where $W_{sc,a}$ is the average wage at age a of the students who attended school s in cohort c, excluding the first promoted manager, and G_{sc} is the age of group sc when a former schoolmate is first promoted to a managerial position.[21] Groups that have not yet experienced a promotion by age a serve as controls. The remaining fixed effects restrict the comparison to groups of the same age who faced the same confounding shocks stemming from local economic

[20] I consider the first promotion taking place in each group, restricting to private-sector firms with at least 25 employees, excluding the public sector and education.

[21] Age is defined at the group level as $a = t - c + 16$, where t is the calendar year and c the enrollment cohort, so that all members of a group share the same age in a given year (86.4% enroll at age 16 or 17). This is a convention motivated by the research design: career evolution is measured by the number of years elapsed from initial peer exposure.

conditions and career-specific wage paths. The fixed effects $\psi_{r(s),c,a}$, for each combination of the school's region (NUTS3) $r(s)$, enrollment cohort c and age a, absorb local economic shocks: because cohort and age together determine the calendar year, they compare groups in the same region in the same year. The fixed effects $\xi_{k(s),a}$, allow wage profiles to differ by high-school track k, effectively comparing students who received a comparable education in high school. The coefficient δ_ℓ measures the effect of a peer's promotion on wages ℓ years after it. Following Callaway and Sant'Anna (2021), I first estimate a separate effect for each treatment age g and age a, $ATT(g, a)$, comparing groups with $G_{sc} = g$ to groups not yet treated at age a, and then aggregate them into δ_ℓ by averaging $ATT(g, g + \ell)$ across treatment ages, weighted by group size.

This design offers at least two advantages. First, exploiting promotions it is unlikely to be confounded by changes in labor demand at the connected firm that would arise from looking at hiring or mass layoff events.[22] Second, under the identifying assumption, differences in individual productivity emerging from interactions in high schools are not confounding the estimated effect. Groups are allowed to have different levels of wages at baseline, while differences in wage growth between groups who experienced promotion at different times are unlikely to capture returns to human capital accumulation, ultimately offering useful variation to identify spillovers from peers' career advancement.

Figure 8 reports the estimates of δ_ℓ for log hourly wages. Before the promotion, wages of treated and not-yet-treated groups evolve in parallel: the coefficients between four and two years before the promotion are close to zero, and only the earliest lead is small and negative. After the promotion, wages of former schoolmates rise steadily, by about 1% five years after the event and 2% eight years after. Averaged over the eight years following the promotion, the effect is 0.9% (0.009 log points, s.e. 0.002).[23] As shown in Figure D.1 (Appendix D), the promotion does not affect employment: the average effect on the probability of being employed is zero and precisely estimated. The wage gains therefore do not reflect changes

[22] One could restate the identifying restriction in the tradition of the study of the identification of peer effects initiated by Manski (1993) and developed in Moffitt (2001); De Giorgi et al. (2010); Bramoullé et al. (2009): identification requires the individual promotion (or any other alternative candidate shock) to affect peers' outcomes only via the promoted manager herself and not directly.

[23] These results are robust to the choice of estimator. The imputation estimator of Borusyak et al. (2024) and the estimator of de Chaisemartin and D'Haultfœuille (2024) yield average effects of 1.1% and 1.4%, against 0.9% for the baseline (Appendix D, Table D.1).

in the composition of employed workers. These results are not driven by workers who follow the promoted manager. Excluding from each group the schoolmates who have worked at the same plant as the promoted manager leaves the estimates essentially unchanged, with an average effect of 0.9% (s.e. 0.002; Appendix D, Figure D.3).

[Figure 8 HERE]

In a model where workers bargain over their wage with their employer, the value of the alternative employment opportunity for their bargaining position is increasing in the wage offered by the competing employer (Postel-Vinay and Robin, 2002). Figure 9 estimates the model separately for groups whose first promoted manager works at a firm in the top 20% of the distribution of AKM firm effects and others. Consistently, the spillover effect is larger when the manager is promoted at a high-paying firm, reaching about 2% after eight years against about 1% for other groups.

[Figure 9 HERE]

Overall, the results in this section highlight how peer influence extends beyond contemporaneous effects developing close to school age. Early developed social ties are the backbone of the social networks workers exploit on the labor market to access job opportunities. Consistently, the effects of exposure on labor market earnings extend beyond what increases in education would predict and emerge in a setting in which peers increase hiring probabilities at connected firms and improve bargaining positions when their careers advance.

6 Mediation Analysis

The previous sections establish two facts. First, high-SES schoolmates increase adult earnings (Section 3). Second, social ties developed in school are connections on the labor market (Section 5). A growing literature shows that parental background is a strong predictor of access to higher wage jobs (Staiger, 2026; Forsberg et al., 2026; San, 2026; Dobbin and Zohar, 2026). As a result, an important component of the value of early peer exposure is the set of job opportunities they grant access to. The goal of this section is to quantify this channel separately.

6.1 Potential Outcomes Framework

A useful way to characterize the estimand of interest is the following: how important is improved access to social networks in determining the effect of peer exposure? In the framework of mediation analysis (Imai et al., 2010; Huber and Frölich, 2020; Schenk, 2025), this is equivalent to asking how different would have been the effect of a treatment (peer exposure) on an outcome (earnings), if the treatment had not affected the mediator (improved access to jobs via social contacts). Crucially, a comparison of earnings among students who had the same peer parental background but whose peers got promoted at different ages would likely compound different channels. For example, even under random assignment into peer groups, a schoolmate who had a successful career might have also served as a good study partner, influencing both human capital and social networks.

I build on recent advances in the mediation literature (Schenk, 2025) to develop a formal framework that allows one to isolate the effect of peer exposure on earnings that operates through the timing of former schoolmates' promotions, exploiting a set of identifying assumptions comparable to those underlying the previous results of this paper. In short, within school across cohort random variation in group composition identifies the effect of peer exposure on both adult earnings (outcome) and schoolmates' promotions' timing (mediator). At the same time, the effect of each one-year-earlier shift in the timing of a peer's promotion is identified comparing students whose first peer promotion occurs one year apart, in a difference in differences design similar to the one used in the previous section. This ef-

fect is identified as the average for all students who experienced a promotion in each year; a portability condition assumes it is the same as the average effect among those whose peer promotion timing was shifted by a change in peer parental background. The mediated effect is then the product of timing shifts and their effects, summed over all ages at which a peer promotion can occur.

I formalize this argument in a potential outcomes model in Appendix E, sketching here its main components. For each student i, let $P_i \in \{r, r'\}$ denote peer exposure. Peer quality is discretized distinguishing between exposure to the top (r', high SES peers) and the bottom (r, low SES peers) tercile of average peer parental earnings. Let S_i denote the age at which a former schoolmate of i is first promoted to manager. Finally, let $Y_{i,a}(p, s)$ be the *potential* earnings of student i at age $a \in [a_0, A]$ if she were exposed to peer quality p and her first peer promotion occurred at age s, and $S_i(p)$ be the *potential* age of her first peer promotion under exposure p. Only realized earnings are observed: $Y_{i,a} = Y_{i,a}(P_i, S_i(P_i))$ and $S_i = S_i(P_i)$.

The target parameter of interest in this application is the *mediated effect*:

$$\theta_a(r, r') = \mathbb{E}\big[Y_{i,a}\big(r', S_i(r')\big) - Y_{i,a}\big(r', S_i(r)\big) \mid P_i = r'\big]. \tag{5}$$

To interpret it, consider a student who attended school with high-SES peers. $\theta_a(r, r')$ compares her actual earnings at age a with the earnings she would have had if she had kept exactly the same schoolmates, but her first former schoolmate had been promoted at the age at which this would have happened had she been exposed to low-SES peers. This is a cross-counterfactual comparison, where all the components of peer exposure are held fixed at the high-exposure level, except for the career evolution of peers, whose timing is set to the one of low-SES peers. The core challenge is that the potential earnings under exposure level r' and peer promotion date $S_i(r)$ are never observed for the population of interest of students who were exposed to high-SES peers.

The target parameter at age a can be reformulated as the combination of two sets of causal effects: the change in cumulative peer promotion rate by each age q due to improved exposure and the effect of anticipating peer promotion from $q + 1$ to q, evaluated at age a,

summing over all potential changes taking place before the relevant age a,[24] as in:

$$\theta_a(r, r') = \sum_{q=a_0}^{a} \pi_c^q(r, r')\, \tau_{a,\text{DiD}}^q. \tag{6}$$

Shift in peer promotion timing. $\pi_c^q(r, r')$ is the effect of exposure on the probability of having experienced a peer promotion by age q. It is identified under the same assumption underlying the main within school across cohort design of this paper (Hoxby, 2000): fluctuations in group composition around school specific time trends are uncorrelated to potential outcomes. I estimate it from a series of separate OLS regressions of the same model as eq. (1), replacing the main outcome with an indicator equal to one for those individuals who experienced a peer promotion by each age q and zero otherwise.

Figure 10 reports the result by showing the realized promotion rate for groups with high peer quality and the predicted counterfactual promotion rate had peer parental background been lower. While early promotion rates are similar across group quality, the gap emerges after age 33 when those who had high-SES peers are around 4 percentage points more likely to have a former schoolmate promoted to manager than those who had low-SES peers.

[Figure 10 HERE]

Effect of an earlier peer promotion. $\tau_{a,\text{DiD}}^q$ is the effect on earnings at age a of experiencing the first peer promotion at age q rather than $q+1$, for students exposed to high-SES peers. I estimate it with a difference-in-differences design that compares the wage growth of students whose first peer promotion occurs at age q with that of students whose first peer promotion occurs one year later.[25] The design is closely related to the one in eq. (4): differences in wage levels between groups are absorbed, and identification rests on parallel trends absent differences in promotion timing. It differs in two respects. First, the estimand is the effect of experiencing the promotion one year earlier, rather than the effect of the promotion itself. Second, the control group is restricted to students whose first peer promotion occurs

[24] This extends the work of Schenk (2025) to a context in which the effect of the treatment is identified within school across cohorts and the mediator is staggered across different periods.

[25] I balance the control group to match the distribution of the treatment group in terms of cohort and school type, to address the same concerns discussed in section 5.

exactly one year later. Because treated and control students experience the promotion only one year apart, they are close in all the determinants of promotion timing, which makes parallel trends more plausible than in a comparison with all students not yet treated.

As shown in eq. (6), I allow for a differential effect of anticipating the promotion by threshold-age q on earnings at each horizon-age a. The mediated effect at each horizon age a is the sum of the effects of shifts in promotion timing $\tau^q_{a,\text{DiD}}$ at each threshold age q until the considered horizon a, weighted by the change in the probability of such shifts due to improved exposure $\pi^q_c(r, r')$. Figure 11 plots the event study estimates of the effect of anticipating a peer promotion from $q+1$ to q on earnings in each year before and after the promotion, aggregating the effects estimated for each potential threshold age q according to the same weights $\pi^q_c(r, r')$. Before the promotion, groups who have a member promoted for the first time one year apart experience the same wage growth, as shown by the absence of differential pre-trends in the graph. After the promotion, the early-promoted group has an additional increase in earnings with respect to the late-promoted group of around \$250 per year, which fades out after the third year.

[Figure 11 HERE]

Finally, Figure 12 reports the main result: for each age between 30 and 40, the total effect of exposure to high-SES peers on earnings, $\beta_a(r, r')$, and the component that operates through earlier promotions of former schoolmates, $\theta_a(r, r')$. The total effect grows in the first years of the career and stabilizes at around \$1,770 per year after age 35. The mediated component follows the same profile: it is close to zero in the early thirties, when promotions are rare in both groups, and rises to around \$230 per year as careers progress. Averaged over the late-career ages marked by the horizontal lines in Figure 12, earlier access to former schoolmates in managerial positions accounts for 13% of the return to peer exposure.[26]

[Figure 12 HERE]

Overall, this section shows that peers' influence extends beyond contemporaneous effects while in school, with important implications for the transmission of opportunities and job

[26]The 90% confidence interval on this share is $[0\%, 33\%]$. Inference is obtained from the empirical distribution of 1,000 Bayesian bootstrap draws, in which each school receives an independent exponential weight with mean one, to account for clustering at the school level.

access. The next Section expands these considerations by discussing them in the context of the results of this paper and of the existing literature.

7 Discussion

The role of social interactions for intergenerational mobility has been a topic of interest in economics for decades. Theoretical contributions (Loury, 1976; Benabou, 1993; Durlauf, 1996) investigated the role of local communities for human capital accumulation, and empirical studies documented the role of local economic conditions and networks for intergenerational mobility (Chetty and Hendren, 2018a,b; Fogli et al., 2026; Chetty et al., 2022a,b). This paper provides empirical evidence from Denmark showing that peer exposure during high school affects adult earnings and intergenerational mobility, especially for high-SES students. Moreover, it shows that access to higher-paying firms is a key mechanism through which these effects operate, beyond the effect of peers in shaping individual productivity previously considered. This section discusses the implication of these findings in the context of the existing literature and of the institutional context considered.

High schools are a compelling setting to study peer exposure on adult outcomes for a number of reasons. First, as opposed to earlier school years, high-school students are more likely to develop their social networks independently from their parents, therefore making changes in the composition of pools of potential friends more likely to affect the social networks they form. In this direction, evidence from the U.S., studying Facebook friendship networks, by Chetty et al. (2022a,b) shows how cross-SES friendships are formed in U.S. high schools, stressing the role of *exposure* in offsetting the natural tendency of creating links within SES clusters. Second, a large literature has shown high-school exposure to be influential for lifetime outcomes. For example, Cattan et al. (2025) documents the role of high-school schoolmates for access to elite university in Norway, while Black et al. (2013) and Brenøe and Zölitz (2020) show how high-school composition affects long-run economic outcomes.

Some specific features of the Danish institutional context make it a particularly interesting setting to study the role of social interactions in shaping intergenerational mobility. First, Danish high schools have little to no tuition fees and are financed from state-level taxation. As a consequence, concerns about school resources shifting with changes in the students' parental backgrounds are less pronounced than in contexts where schools are financed by

tuitions or local taxation. At the same time, the Danish high-school system is characterized by a relatively low level of segregation by parental earnings, especially for low and middle parental earnings students. On the one hand, this offers variation in school composition that allows to identify the causal effect of peer exposure on adult outcomes. On the other hand, it makes the overall effect of segregation on intergenerational mobility smaller than in more segregated contexts, such as the US. More research is needed to assess the external validity of the causal effect of peer exposure in more segregated contexts. However, assuming it has similar effects, its contribution to intergenerational persistence would be even larger in more segregated contexts.

A central limitation of this analysis is that social networks are not directly observed. While the endogenous formation of social ties is not a concern for identification under the assumptions stated in the paper, it is important to qualify the interpretation of the results. The treatment effect identified in this paper is best interpreted as an intention-to-treat effect, where the treatment is a change in the cost of forming social ties with higher-SES peers due to a change in the availability of such peers. However, two considerations are worth noting. First, changes in exposure are a policy relevant margin, as policy-makers are often interested in the effect of desegregation policies on social mobility. Second, the results on the role of social contacts on the labor markets show that connections developed in high school are influential for job access of low-SES students connected to high wage job opportunities by high-SES peers.

Linking school exposure to complete employment trajectories makes it possible to study directly how school ties operate on the labor market. In doing so, this paper joins a small literature on the role of schoolmates in job access (Eliason et al., 2023), and complements a broader range of evidence on ties formed later in life, among college peers (Zimmerman, 2019; Zhu, 2025; Fischer et al., 2026) and coworkers (Hensvik and Skans, 2016; Glitz and Vejlin, 2021; Saygin et al., 2021; Caldwell and Harmon, 2019). The results show that ties formed as early as high school remain valuable well into workers' careers, strengthening the case for social ties as a determinant of access to jobs (Calvó-Armengol and Jackson, 2007; Bolte et al., 2026). Finally, my results are similar to those of Eliason et al. (2023) even though the two designs exploit alternative comparisons: I compare actually and almost connected

destinations for the same worker, while they compare connected and unconnected workers for the same origin–destination pair of firms. As the two designs rest on different identifying assumptions, the similarity in our results further strengthens the case on the value of early developed ties for access to jobs.

The implications of these results are best appreciated in the context of a growing literature on the transmission of employers and access to high-wage jobs from parents to their children. Children often inherit occupations and employers from their parents (Kramarz and Skans, 2014; Staiger, 2026; Ventura, 2024; San, 2026), and those from higher-SES families are more likely to work at higher-paying firms (Dobbin and Zohar, 2026; Forsberg et al., 2026). When high-SES students have privileged access to high-wage jobs, ties with them can themselves become a channel of job access. The results of this paper support this view: access to good jobs is transmitted not only vertically, from parents to children, but also horizontally, across peers.

Job access translates in wage gains. The findings of this paper are in line with models of the labor market where social ties shape outside options (Caldwell and Harmon, 2019), ultimately affecting the worker's bargaining position with their employer (Postel-Vinay and Robin, 2002). In particular the results on wage spillovers from peer's career improvements are consistent with earlier research from Bagger et al. (2014), who estimate a structural model of job search and human capital accumulation and show that outside options are key for Danish workers to sustain wage growth early in their careers.

The last section of this paper builds on recent advances in mediation analysis (Imai et al., 2010; Huber and Frölich, 2020; Schenk, 2025) to isolate the share of the return to exposure that operates through improved access to jobs via higher-SES peers. Distinguishing mechanisms is particularly relevant for the study of peer effects, as they shape outcomes through multiple channels, such as social norms (Bursztyn et al., 2018; Genicot and Ray, 2020), information (Bellue, 2023; Cattan et al., 2025), and human capital (Fruehwirth and Gagete-Miranda, 2019) or rank concerns (Tincani, 2026), each with different implications for policy.

Moreover, the finding that peer career advancement mediates the effect of schoolmate exposure is central to the question of this paper: peer exposure is not only about the develop-

ment of individual skills and productivity, the focus of seminal models of social interactions and inequality (Loury, 1976; Benabou, 1993; Durlauf, 1996; Fogli et al., 2026), but also about access to opportunities in labor markets where information is asymmetric. Peers are more than learning partners: they are labor-market matchmakers.

Two main conclusions follow. First, exposure matters for lifetime outcomes and, especially at the top of the parental earnings distribution, is an important driver of the transmission of economic inequality across generations. A policy maker seeking to foster mobility by facilitating interactions across parental backgrounds should consider the endogenous responses of high-SES families to increased exposure to low-SES peers, but the results of this paper show that the channel such a policy would act on is quantitatively relevant. Second, peer exposure shapes the networks that match workers to firms, with potential consequences for aggregate efficiency (Calvó-Armengol and Jackson, 2007; Bolte et al., 2026). These consequences should be accounted for when assessing the equity–efficiency trade-offs of desegregation policies, as well as of alternative technologies that provide information on the labor market, such as screening tools or job platforms.

8 Conclusions

This paper shows that differences in parental resources extend beyond the family through the peers children grow up with. A $1 increase in the average parental earnings of schoolmates raises annual adult earnings by approximately $0.10. The effect of exposure on intergenerational mobility increases with segregation, and it accounts for 3.6% of the transmission of earnings inequalities in Denmark, while it would explain 11% under US residential sorting by parental earnings. The effect operates beyond education: school ties persist in the labor market, lead to hires at the same firms, and raise wages as former schoolmates advance in their careers. A formal mediation analysis shows that earlier access to schoolmates in managerial positions alone accounts for 13% of the return to peer exposure. Peers are therefore not only learning partners, as in the classic models of social interactions and inequality, but also labor-market matchmakers: segregation reproduces advantage across generations by shaping who children know when they enter the labor market, as well as what they learn.

Tables

Table 1: Descriptive Statistics

	Parental Earnings			
	Low	Mid	High	All
Panel A: Children				
Female	0.60	0.59	0.54	0.58
	(0.49)	(0.49)	(0.50)	(0.49)
Earnings (USD)	53,570.54	59,439.75	67,318.08	60,043.86
	(35,122.88)	(34,346.71)	(53,275.02)	(42,156.14)
Earnings (rank)	44.82	50.48	56.26	50.46
	(28.03)	(27.82)	(29.77)	(28.93)
College	0.64	0.70	0.78	0.70
	(0.48)	(0.46)	(0.42)	(0.46)
Panel B: Parents				
Earnings (USD)	27,280.54	46,295.40	72,458.64	48,463.54
	(10,939.39)	(4,403.06)	(36,003.90)	(28,617.37)
Earnings (rank)	17.50	51.00	84.00	50.50
	(9.81)	(9.52)	(9.52)	(28.87)
College	0.31	0.46	0.71	0.49
	(0.46)	(0.50)	(0.45)	(0.50)
Panel C: Peers' Parents				
Earnings (USD)	46,579.49	47,649.66	51,218.77	48,463.54
	(7,059.68)	(7,027.57)	(9,371.68)	(8,133.63)
Earnings (rank)	47.99	49.62	53.95	50.50
	(8.38)	(8.41)	(9.83)	(9.24)
College	0.46	0.47	0.54	0.49
	(0.16)	(0.16)	(0.17)	(0.17)
Group Size	98.56	100.12	98.41	99.03
	(63.67)	(61.75)	(55.98)	(60.59)
Observations	139,739	135,624	135,617	410,980

Note: The table presents summary statistics for the main sample of students enrolled at Danish high schools from 1988 to 2000 and their parents, by tercile of parental earnings, including the sample mean and standard deviation (in parentheses).

Table 2: Variation in School Composition

	(A) Sample Averages				(B) Residual SD		
	Parental Earnings						
	Low	Mid	High	All	(1)	(2)	(3)
Peers' Par. Earnings (rank)	47.99	49.62	53.95	50.50	3.70	3.58	3.20
	(8.38)	(8.41)	(9.83)	(9.24)	[40%]	[39%]	[35%]
Peers' Par. Earnings (USD)	46,579.49	47,649.66	51,218.77	48,463.54	3,611.43	3,332.55	2,960.71
	(7,059.68)	(7,027.57)	(9,371.68)	(8,133.63)	[44%]	[41%]	[36%]
% Peers Low T.	37.93	34.21	29.75	34.00	6.73	6.15	5.27
	(11.21)	(10.80)	(11.15)	(11.55)	[56%]	[51%]	[44%]
% Peers Mid T.	33.22	34.86	30.96	33.02	5.90	5.79	5.16
	(7.30)	(7.23)	(8.32)	(7.79)	[71%]	[70%]	[63%]
% Peers Top T.	28.92	30.97	39.31	33.03	6.40	5.36	4.96
	(12.57)	(12.90)	(15.44)	(14.41)	[43%]	[36%]	[33%]
School Size	98.56	100.12	98.41	99.03			
	(63.67)	(61.75)	(55.98)	(60.59)			
School FEs					✓	✓	✓
Cohort FEs						✓	✓
School T. Trends							✓
Observations	139,739	135,624	135,617	410,980	410,980	410,980	410,980

Note: The left panel reports sample averages of schoolmates' characteristics by tercile of own parental earnings and overall, with standard deviations in parentheses. The right panel reports the standard deviation of the residuals from regressing each variable on progressively richer sets of fixed effects; square brackets report each residual SD as a percentage of the observed standard deviation in the full sample (column All).

Table 3: Serial Correlation in School Composition

	Observed	Residualized		
	(1)	(2)	(3)	(4)
Significant at 10%	16.98% [81]	9.85% [47]	10.27% [49]	9.43% [45]
Significant at 5%	9.01% [43]	5.45% [26]	6.08% [29]	3.77% [18]
Significant at 1%	2.94% [14]	1.89% [9]	1.68% [8]	0.42% [2]
Cohort FEs		✓	✓	✓
School FEs			✓	✓
School T. Trends				✓
Schools	477	477	477	477

Note: The table reports the share of schools in which the correlation over time in schoolmates' parental earnings is statistically different from zero at the indicated significance level; the number of schools is in square brackets. Column (1) measures school composition as observed in the data; columns (2)–(4) measure it as the residual from a linear model including the fixed effects indicated at the bottom of the table.

Table 4: Balance Test: Schoolmates' Parental Earnings and Predetermined Characteristics

	Student			Mothers			Fathers		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
First Born	0.032	0.001	0.001						
	(0.004)	(0.001)	(0.001)						
	[0.000]	[0.319]	[0.329]						
Only Child	0.035	-0.000	-0.000						
	(0.006)	(0.001)	(0.001)						
	[0.000]	[0.611]	[0.698]						
Par. Earnings (USD)	0.239	0.008	-0.001						
	(0.025)	(0.002)	(0.001)						
	[0.000]	[0.000]	[0.352]						
High School				-0.070	-0.000	0.000	-0.112	-0.000	-0.000
				(0.009)	(0.001)	(0.001)	(0.013)	(0.001)	(0.001)
				[0.000]	[0.717]	[0.775]	[0.000]	[0.963]	[0.580]
Missing Parental ID				-0.004	-0.001	-0.001	0.001	-0.003	-0.001
				(0.002)	(0.001)	(0.001)	(0.003)	(0.001)	(0.001)
				[0.047]	[0.177]	[0.301]	[0.665]	[0.003]	[0.083]
Missing Education				0.008	-0.001	0.000	0.006	-0.002	-0.001
				(0.004)	(0.001)	(0.001)	(0.004)	(0.001)	(0.001)
				[0.024]	[0.431]	[0.562]	[0.138]	[0.031]	[0.345]
Managerial Position				0.033	0.002	0.000	0.070	0.003	0.000
				(0.003)	(0.001)	(0.001)	(0.008)	(0.001)	(0.001)
				[0.000]	[0.022]	[0.996]	[0.000]	[0.011]	[0.788]
Full-time Work				0.054	0.003	0.000	0.029	0.004	0.001
				(0.005)	(0.001)	(0.001)	(0.007)	(0.001)	(0.001)
				[0.000]	[0.000]	[0.665]	[0.000]	[0.000]	[0.022]
Hourly Wage				0.127	0.004	0.002	0.187	0.003	0.001
				(0.016)	(0.001)	(0.001)	(0.019)	(0.001)	(0.001)
				[0.000]	[0.000]	[0.022]	[0.000]	[0.001]	[0.141]
College				0.193	0.003	0.001	0.227	0.003	0.001
				(0.017)	(0.001)	(0.001)	(0.021)	(0.001)	(0.001)
				[0.000]	[0.000]	[0.163]	[0.000]	[0.001]	[0.043]
Earnings				0.195	0.009	0.003	0.191	0.005	-0.000
				(0.014)	(0.001)	(0.001)	(0.025)	(0.002)	(0.001)
				[0.000]	[0.000]	[0.001]	[0.000]	[0.017]	[0.779]
School FEs		✓	✓		✓	✓		✓	✓
Cohort FEs		✓	✓		✓	✓		✓	✓
School T. Trends			✓			✓			✓

Note: The table reports the estimates plotted in Figure 2. Each cell reports the coefficient from a separate regression of schoolmates' parental earnings on the predetermined characteristic indicated in the row, for the student (Student columns) or for the student's mother or father (Mothers and Fathers columns). Standard errors are clustered at the school level and reported in parentheses, p-values are reported in square brackets. Column (1) measures school composition as observed in the data; columns (2) and (3) measure it as the residual from a linear model including the fixed effects indicated at the bottom of the table. All regressions control for the school-level average of parental earnings to account for the mechanical negative correlation induced by the leave-one-out measure (Guryan et al., 2009).

Table 5: Peer Exposure and Adult Earnings

	2015 USD				Ranks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Par. Earnings	0.249***	0.230***	0.219***	0.218***	0.160***	0.138***	0.131***	0.130***
	(0.034)	(0.037)	(0.038)	(0.038)	(0.003)	(0.002)	(0.002)	(0.002)
Peers' Par. Earnings		0.307***	0.123***	0.096***		0.248***	0.062***	0.046***
		(0.043)	(0.029)	(0.028)		(0.018)	(0.016)	(0.015)
Cohort FEs			✓	✓			✓	✓
School FEs			✓	✓			✓	✓
School T. Trends				✓				✓
Observations	410980	410980	410980	410980	410980	410980	410980	410980
R^2	0.09	0.09	0.11	0.12	0.12	0.13	0.16	0.16

Note: The table reports OLS estimates of eq. (1). The dependent variable is adult earnings, measured in 2015 USD in columns (1)–(4) and in percentile ranks within birth cohort in columns (5)–(8); parental earnings and schoolmates' parental earnings are measured in the corresponding units. All specifications control for gender and year of birth. Fixed effects are indicated at the bottom of the table. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Heterogeneity in the Effect of Peer Exposure

	(1)	(2)	(3)	(4)	(5)
Peers' Par. Earnings	0.118**	0.044	0.066*	0.110**	0.088***
	(0.051)	(0.029)	(0.039)	(0.054)	(0.030)
× Female	-0.038				
	(0.063)				
× College Edu. Parents		0.089***			
		(0.026)			
× Mid Par. Earnings			0.017		
			(0.024)		
× Top Par. Earnings			0.062		
			(0.045)		
× ≥ 60 Peers				-0.034	
				(0.075)	
× Urban School					0.024
					(0.075)
Cohort × Group FEs	✓	✓	✓	✓	✓
School FEs	✓	✓	✓	✓	✓
School T. Trends	✓	✓	✓	✓	✓
Observations	410980	410980	410980	410980	410980
R^2	0.12	0.12	0.12	0.12	0.12

Note: The table reports estimates of eq. (1) in which schoolmates' parental earnings are interacted with indicators for: female students (column 1), students with at least one college-educated parent (column 2), the middle and top terciles of own parental earnings, with the bottom tercile as the reference group (column 3), school cohorts with at least 60 peers (column 4), and urban schools (column 5). The main coefficient reports the effect for the reference group. The dependent variable is adult earnings in 2015 USD. All specifications include group-by-cohort fixed effects, school fixed effects and school-specific time trends. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Estimates of Sorting Across Peer Groups

	Target $\rho = \frac{Cov(\bar{X}_{-i}, X_i)}{Var(X_i)}$	Estimate $\hat{\rho} = \frac{Var(\mu_g)}{Var(X_i)}$	Bounds on ρ $[\hat{\rho} - b,\ \hat{\rho}]$	Groups (mean size)
DK, school × cohort	0.089	0.102	[0.059, 0.102]	5,757 (71)
DK, Neighborhood	0.102	0.107	[0.091, 0.107]	2,141 (192)
U.S., Census tracts	–	0.282	[0.277, 0.282]	72,014 (539)

Note: The first column reports the OLS estimate of ρ for the relevant population, where available. The second column reports $\hat{\rho}$, defined as the ratio of the variance of group means to the total variance in that population. The third column reports exact bounds for ρ as a function of $\hat{\rho}$ accounting for the fact that leaving oneself out mechanically lowers the slope as derived in Appendix B. The last column reports the number of peer groups and, in parentheses, the average number of children per group. Population and peer groups differ across rows and are defined as follows. DK, school × cohort: the sample of Danish high-school students used throughout the paper, with peer groups defined by enrollment in the same school and year, as in the main analysis. DK, Neighborhood: the sample of Danish high-school students used throughout the paper, with peer groups defined by residence in the same municipality at age 16 (residential sorting). U.S., Census tracts: children born in 1978–83, with peer groups defined by the census tract in which they grew up, pooling birth cohorts (Opportunity Atlas). A direct estimate of ρ is unavailable for the U.S.

Table 8: Education and Labor Market Outcomes by Parental Earnings

	Par. Earnings Tercile			
	Low	Mid	Top	Total
High School	0.965	0.982	0.986	0.977
College	0.660	0.721	0.794	0.725
Pred. Wage/h: Level × Field	35.400	36.170	37.950	36.486
	[131,436]	[126,879]	[124,732]	[383,047]
Employed	0.934	0.956	0.950	0.946
Wage/h	32.445	34.283	37.896	34.862
	[130,544]	[129,629]	[128,775]	[388,948]
Firm AKM	-0.006	0.007	0.036	0.012
	[125,920]	[125,641]	[124,404]	[375,965]
Firm AKM - Top Tercile	0.363	0.403	0.477	0.414
Firm AKM - Top Quartile	0.260	0.297	0.369	0.308
Firm AKM - Top Decile	0.068	0.081	0.121	0.090
Observations	139,739	135,624	135,617	410,980

Note: The table reports average education and labor market outcomes at ages 35–39 by tercile of own parental earnings (columns Low, Mid and Top) and for the full sample (column Total). High School and College are indicators for having completed a high-school and a college degree. Pred. Wage/h: Level × Field is the average hourly wage (2015 USD) at ages 35–39 among individuals with the same completed education level and 1-digit field of education, computed on individuals outside the estimation sample (cohorts 1960–1969). Employed is an indicator for having ever been employed between ages 35 and 39. Wage/h is the hourly wage in 2015 USD, averaged over ages 35–39 conditional on employment. Firm AKM is the plant-specific pay premium, estimated as the plant fixed effect from an AKM decomposition of wages (Abowd et al., 1999), averaged over ages 35–39. Firm AKM – Top Tercile, Top Quartile and Top Decile are indicators for working at a plant whose pay premium lies in the top tercile, quartile and decile of the distribution of plant pay premia; they are defined for the full sample and equal zero for individuals not employed in that period.

Table 9: Peer Exposure, Education and Labor Market Outcomes

	Labor Market							Education		
	Earnings	Employed	Wage/h	Firm AKM	Top Tercile Firm AKM	Top Quartile Firm AKM	Top Decile Firm AKM	High School	College	Pred. Wage/h: Level × Field
Panel A										
SMs P. Earn.	779.000***	0.001	0.311***	0.002***	0.004*	0.005**	0.003**	0.002***	0.004**	0.125***
	(233.275)	(0.001)	(0.099)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.033)
Observations	410980	410980	388948	375965	410980	410980	410980	410980	410980	383047
R^2	0.12	0.01	0.09	0.10	0.08	0.08	0.04	0.02	0.09	0.13
Mean D.V.	60043.86	.95	34.86	.01	.41	.31	.09	.98	.72	36.49
Effect Size	1.3 %	.13 %	.89 %	15.36 %	.98 %	1.76 %	3.41 %	.23 %	.6 %	.34 %
Panel B										
× Low P. Earn.	535.271*	-0.000	0.230*	0.002**	0.004	0.005*	0.001	0.003***	0.005**	0.085**
	(318.071)	(0.001)	(0.124)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.038)
× Mid P. Earn.	674.697**	0.001	0.104	0.000	0.003	0.004	0.001	0.001*	0.002	0.048
	(304.805)	(0.001)	(0.099)	(0.001)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.037)
× Top P. Earn.	1036.310***	0.002	0.504***	0.003***	0.004*	0.006***	0.006***	0.002**	0.003*	0.194***
	(285.813)	(0.001)	(0.105)	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.037)
Observations	410980	410980	388948	375965	410980	410980	410980	410980	410980	383047
R^2	0.12	0.01	0.09	0.11	0.09	0.09	0.04	0.02	0.10	0.14
Mean D.V., Low P. Earn.	53570.54	.93	32.45	-.01	.36	.26	.07	.96	.66	35.4
Effect Size	1 %	-.02 %	.71 %	-36.14 %	1.05 %	1.79 %	2.17 %	.32 %	.81 %	.24 %
Mean D.V., Mid P. Earn.	59439.75	.96	34.28	.01	.4	.3	.08	.98	.72	36.17
Effect Size	1.14 %	.14 %	.3 %	6.55 %	.71 %	1.32 %	1.15 %	.13 %	.26 %	.13 %
Mean D.V., Top P. Earn.	67318.08	.95	37.9	.04	.48	.37	.12	.99	.79	37.95
Effect Size	1.54 %	.17 %	1.33 %	7.41 %	.85 %	1.68 %	4.66 %	.17 %	.42 %	.51 %

Note: The table reports OLS estimates of eq. (1), with the dependent variable indicated at the top of each column, measured at ages 35–39 and defined as in Table 8. Earnings are annual earnings and Wage/h is the hourly wage, both in 2015 USD. Schoolmates' parental earnings are standardized to have mean zero and standard deviation one. Panel A reports the average effect; Panel B interacts schoolmates' parental earnings with indicators for terciles of own parental earnings. Effect Size reports the coefficient as a percentage of the mean of the dependent variable (in Panel B, of the tercile-specific mean). All specifications control for gender and year of birth and include school fixed effects and school-specific time trends. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 10: Connected Hires

		Good Job		Rich Peer	
	All	Rich Mover	Poor Mover	Rich Mover	Poor Mover
Connected	0.0346***	0.0077***	0.0066***		
	(0.0013)	(0.0006)	(0.0004)		
Almost Connected	0.0503***	0.0200***	0.0148***		
	(0.0014)	(0.0008)	(0.0004)		
Connected, by rich peer				0.0312***	0.0231***
				(0.0022)	(0.0017)
Almost Connected, by rich peer				0.0762***	0.0640***
				(0.0020)	(0.0018)
Connected, by poor peer				0.0308***	0.0317***
				(0.0017)	(0.0015)
Almost Connected, by poor peer				0.0567***	0.0521***
				(0.0019)	(0.0017)
Constant	0.0005***	0.0002***	0.0002***	0.0005***	0.0005***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Increase in P (%)	68.06 %	37.89 %	43.89 %		
	(1.98)	(2.93)	(2.94)		
Increase in P (%), rich peer				40.71 %	35.91 %
				(2.69)	(2.40)
Increase in P (%), poor peer				53.91 %	60.39 %
				(2.97)	(2.53)
N pairs (bn)	265.8	95.9	169.8	95.9	169.8
N students	388,117	136,910	251,207	136,910	251,207
N hiring firms	184,677	35,798	48,301	102,715	155,129

Note: The table reports estimates of eq. (3) on individual–firm pairs, pooling hires at ages 25–39. Connected indicates that the firm employs an actual schoolmate of the individual; Almost Connected indicates that it employs an actual or almost schoolmate (same school, one cohort before or after). Coefficients are in percentage points. Column All includes all hires. In the Good Job columns, the outcome is a hire at a firm in the top tercile of the distribution of plant pay premia (AKM). Rich Peer columns split connections by whether the connected (or almost connected) schoolmate has parents in the top tercile of parental earnings (rich peer) or not (poor peer). Rich Mover and Poor Mover refer to movers with own parental earnings in the top tercile and in the bottom two terciles, respectively. Increase in P reports the percentage increase in the hiring probability at a connected firm relative to a firm with only almost connections, $\hat{\delta}_1/(\hat{\alpha}+\hat{\delta}_2)$. Specifications do not include individual fixed effects; estimates with individual or individual × age fixed effects are unchanged and available upon request. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

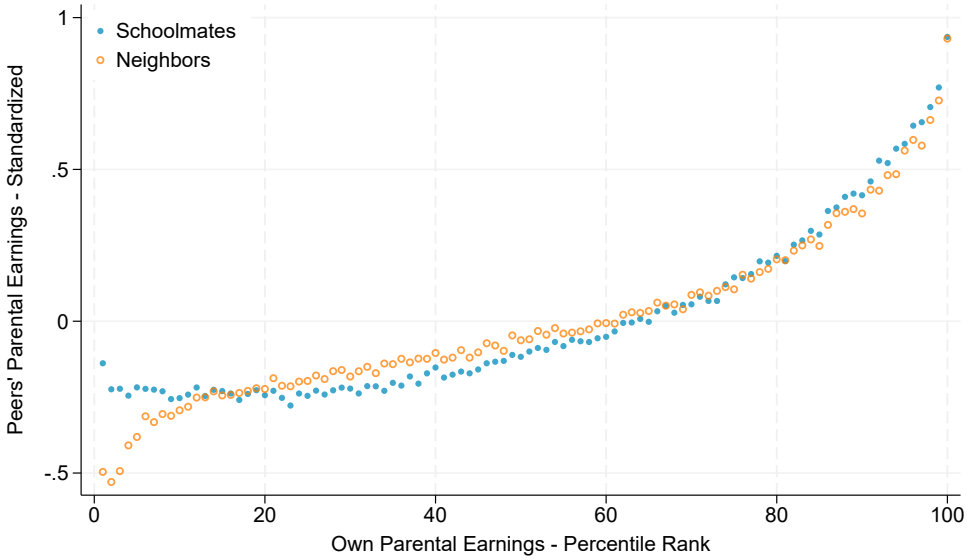
Table 11: Connected Hires by Age

Age	Connected		Almost Connected		Increase in P (%)		Movers	Hiring Firms
25	0.0341***	(0.0038)	0.0689***	(0.0028)	49.04%	(5.83)	84,117	38,241
26	0.0402***	(0.0022)	0.0578***	(0.0020)	68.89%	(3.96)	139,773	50,072
27	0.0391***	(0.0020)	0.0519***	(0.0015)	74.48%	(3.76)	132,457	48,388
28	0.0336***	(0.0020)	0.0528***	(0.0016)	63.02%	(3.99)	124,094	46,680
29	0.0380***	(0.0021)	0.0508***	(0.0015)	73.96%	(4.00)	115,485	44,424
30	0.0403***	(0.0021)	0.0496***	(0.0016)	80.32%	(4.29)	108,108	42,601
31	0.0352***	(0.0020)	0.0479***	(0.0016)	72.64%	(4.14)	102,638	41,286
32	0.0333***	(0.0020)	0.0492***	(0.0015)	67.02%	(4.06)	99,144	40,619
33	0.0323***	(0.0018)	0.0481***	(0.0016)	66.47%	(4.02)	95,885	39,252
34	0.0302***	(0.0019)	0.0479***	(0.0015)	62.43%	(4.05)	92,340	38,224
35	0.0342***	(0.0019)	0.0468***	(0.0015)	72.24%	(4.14)	89,398	37,433
36	0.0313***	(0.0020)	0.0462***	(0.0015)	67.03%	(4.30)	84,856	36,380
37	0.0293***	(0.0019)	0.0455***	(0.0014)	63.63%	(4.16)	80,498	35,323
38	0.0290***	(0.0021)	0.0486***	(0.0016)	59.07%	(4.32)	70,198	31,781
39	0.0346***	(0.0043)	0.0656***	(0.0023)	52.40%	(7.46)	34,076	17,269

Note: The table reports estimates of eq. (3) separately by the individual's age at hire. Connected, Almost Connected and Increase in P are defined as in Table 10. Individuals and Hiring Firms report the number of individuals hired and of hiring firms at each age. Specifications do not include individual fixed effects; estimates with individual fixed effects are unchanged and available upon request. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

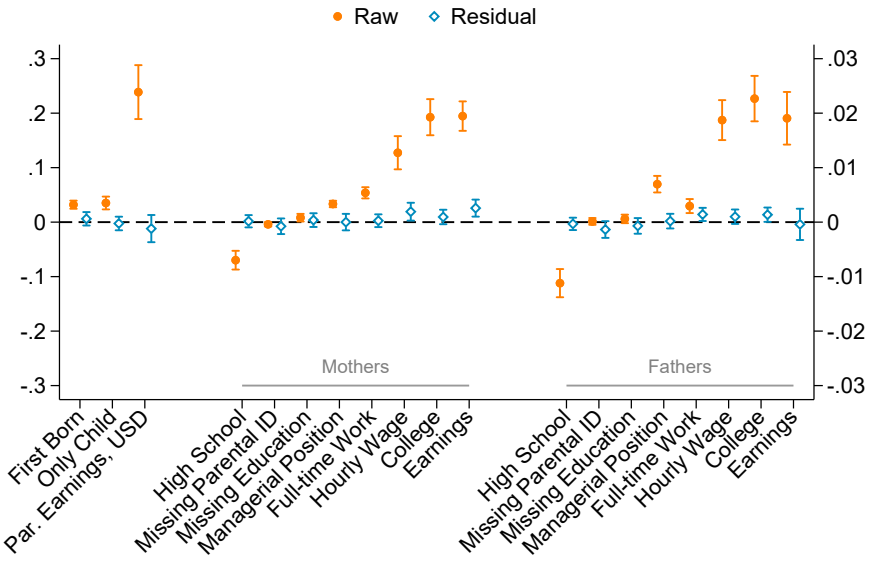
Figures

Figure 1: Sorting across Schools and Neighborhoods



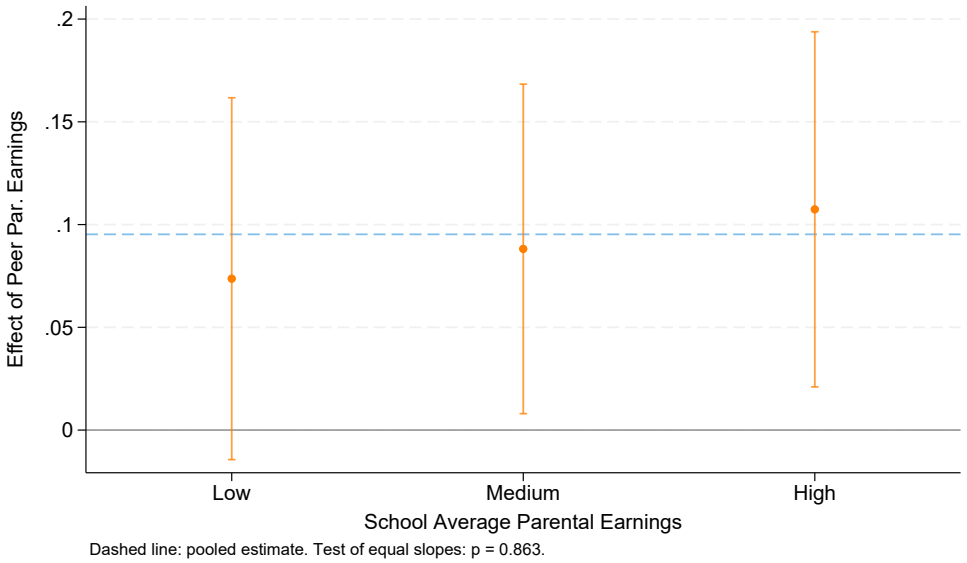
Note: The graph plots the sample average of schoolmates' and neighbors' parental earnings, conditional on the percentile of own parental earnings. Peers' parental earnings are measured as the standardized average of schoolmates' (blue markers) and neighbors' (orange markers) parental earnings. Neighbors are defined as individuals born in the same calendar year and registered as living in the same municipality at the age of 16.

Figure 2: Peer Quality and Individual Characteristics



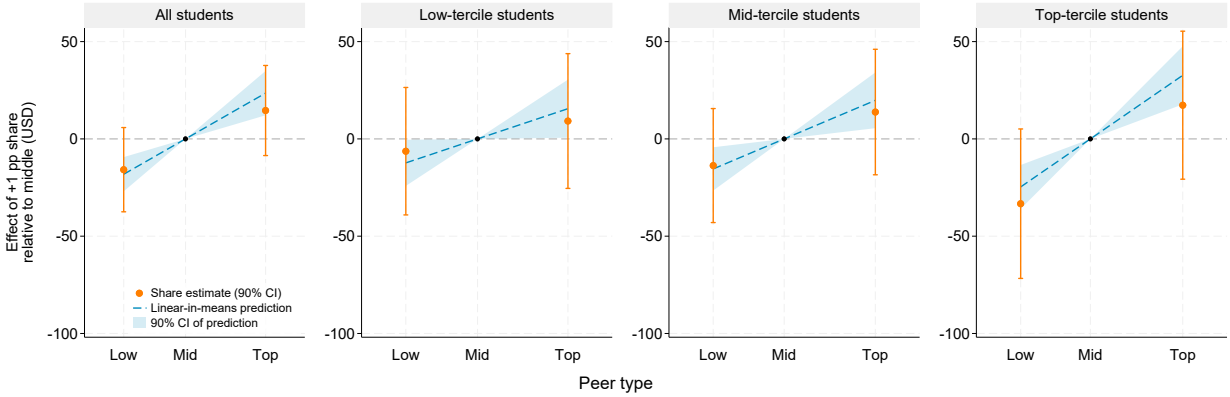
Note: The graph plots coefficients and 95% confidence intervals from separate regressions of the leave-one-out average of schoolmates' parental earnings on each predetermined characteristic listed on the horizontal axis: own characteristics (first three), and characteristics of the student's mother and father. Orange circles use school composition as observed in the data and their magnitude is reported on the left axis. Blue diamonds use school composition as the residual from school fixed effects, cohort fixed effects and school-specific time trends and their magnitude is reported on the right axis. Characteristics are standardized, and all regressions control for the school-level average of parental earnings to account for the mechanical negative correlation induced by the leave-one-out measure (Guryan et al., 2009). Standard errors are clustered at the school level.

Figure 3: Effect of Peer Exposure by School Composition



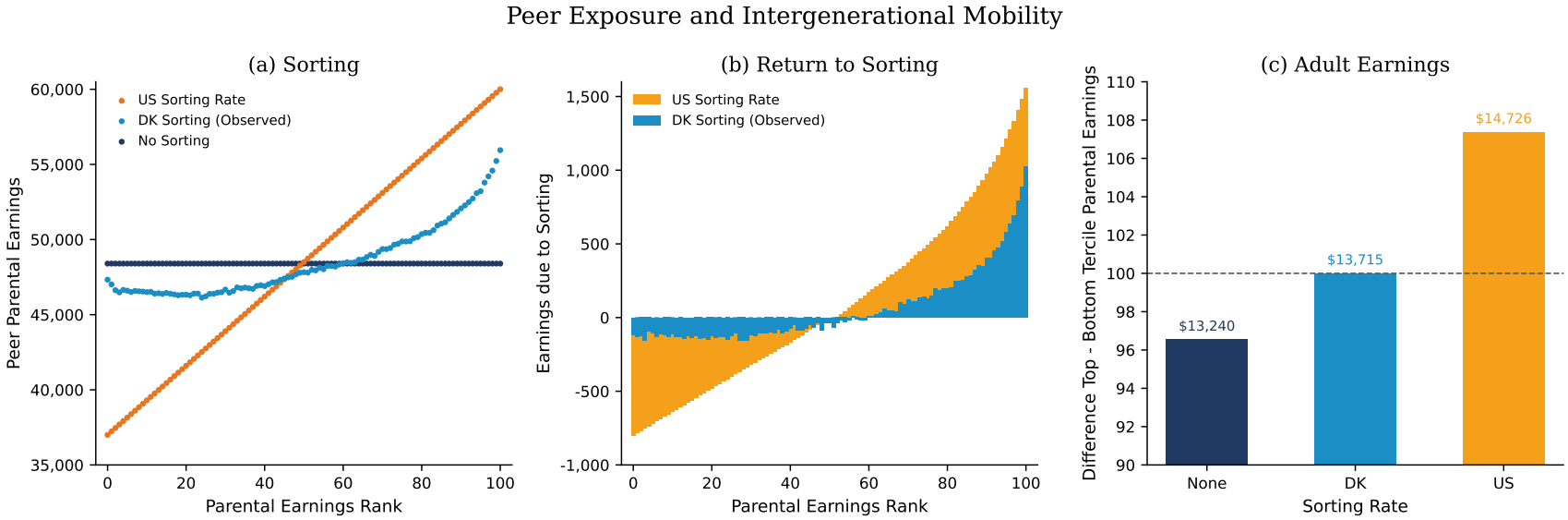
Note: The graph plots the estimated effect of schoolmates' parental earnings on adult earnings (2015 USD), and confidence intervals, separately for schools with low, medium and high average parental earnings. The dashed line reports the pooled estimate (Table 5, column 4). All specifications include school fixed effects, cohort fixed effects and school-specific time trends. Standard errors are clustered at the school level.

Figure 4: Peer Composition and the Linear-in-Means Prediction



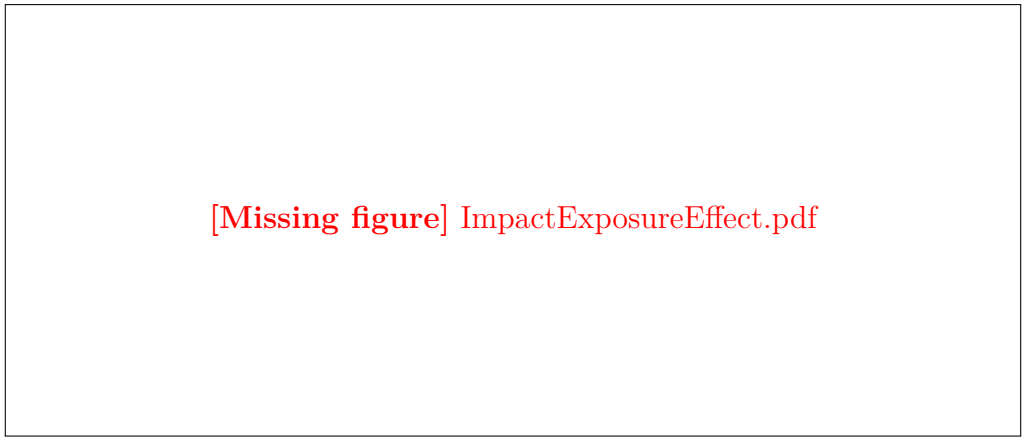
Note: Orange dots report the effect on adult earnings (2015 USD) of a one-percentage-point increase in the share of schoolmates from the bottom (Low) or top (Top) tercile of parental earnings, relative to the middle tercile (Mid), with 90% confidence intervals. The dashed line reports the effect implied by the linear-in-means model, $\hat{\beta}\,(\bar{X}_t - \bar{X}_{Mid})/100$, where $\bar{X}_t$ is the average parental earnings of schoolmates of type t and $\hat{\beta}$ is the estimated effect of schoolmates' parental earnings; the shaded area reports its 90% confidence interval. Peer types are placed on the horizontal axis at their distance in average parental earnings from the middle tercile. The first panel pools all students; the other panels estimate effects and distances separately by tercile of own parental earnings. All specifications include school fixed effects, cohort fixed effects, school-specific time trends and controls for own parental earnings. Standard errors are clustered at the school level.

Figure 5: Peer Exposure and Intergenerational Mobility



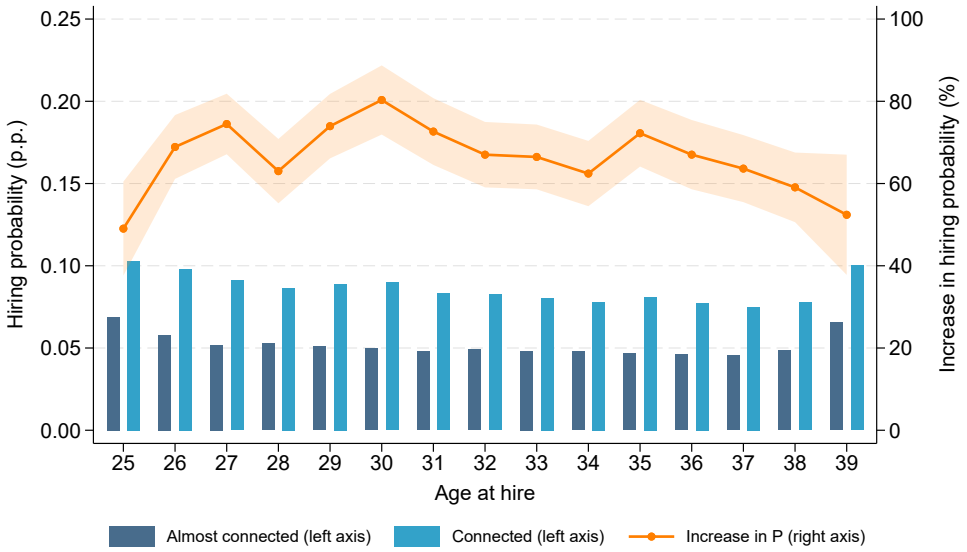
Notes: Panel (a) plots the average schoolmates' parental earnings (leave-one-out mean) against the child's own parental earnings rank, under three scenarios. *No Sorting* assigns every child the population mean. *DK Sorting* is the observed Danish allocation. *US Sorting* rescales exposure to match the degree of residential segregation by parental income across US census tracts (author's calculation based on publicly available data from the Opportunity Atlas Chetty et al. (2026); see Appendix B for the method). Panel (b) shows the change in a child's adult earnings, relative to the no sorting benchmark, implied by the estimated peer effect under each sorting scenario. Panel (c) reports the difference in adult earnings between children of top and bottom tercile parents under each scenario, normalized to the observed Danish level (= 100); levels are shown above the bars. The difference is 11.2% larger in the US scenario and 3.6% in the DK scenario than in the no sorting benchmark, respectively.

Figure 6: Return to Peer Exposure by Parental Earnings



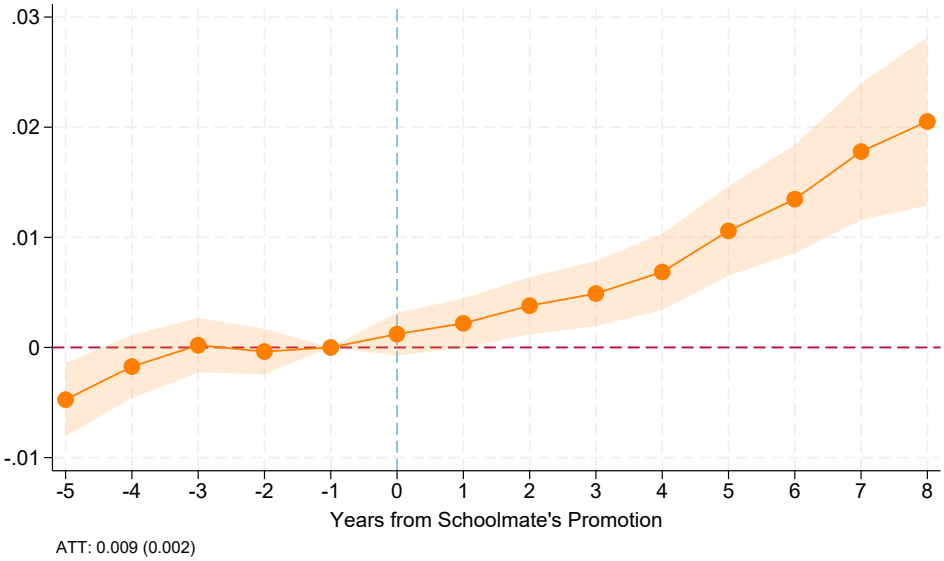
Note: The figure plots the effect of schoolmates' parental earnings on adult earnings (2015 USD) by percentile of own parental earnings. Estimates come from a version of eq. (1) in which the effect of schoolmates' parental earnings is allowed to vary quadratically in own parental earnings rank; these are the percentile-specific returns used in panel (b) of Figure 5.

Figure 7: Connected Hires by Age



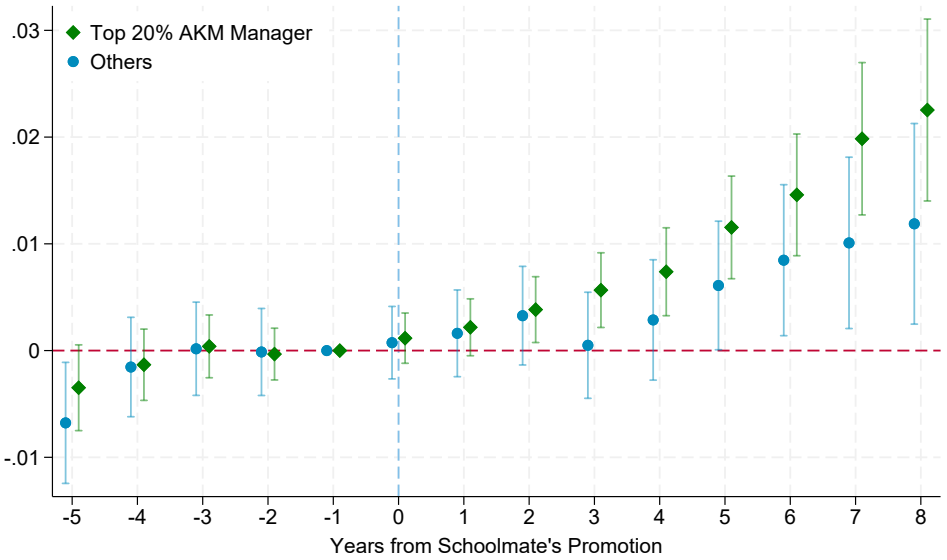
Note: The graph plots the percentage increase in the probability of being hired by a firm employing an actual schoolmate, relative to a firm employing only almost schoolmates, separately by age at hire (Table 11), with 95% confidence intervals (orange, left axis). Bars report the estimated hiring probability, in percentage points (right axis), at firms employing only almost schoolmates ($\hat{\alpha}+\hat{\delta}_2$, dark blue) and at firms employing an actual schoolmate ($\hat{\alpha}+\hat{\delta}_1+\hat{\delta}_2$, light blue). Estimates come from eq. (3) without individual fixed effects, estimated separately by age; estimates with individual fixed effects are unchanged and available upon request.

Figure 8: Schoolmate's Promotion and Wages



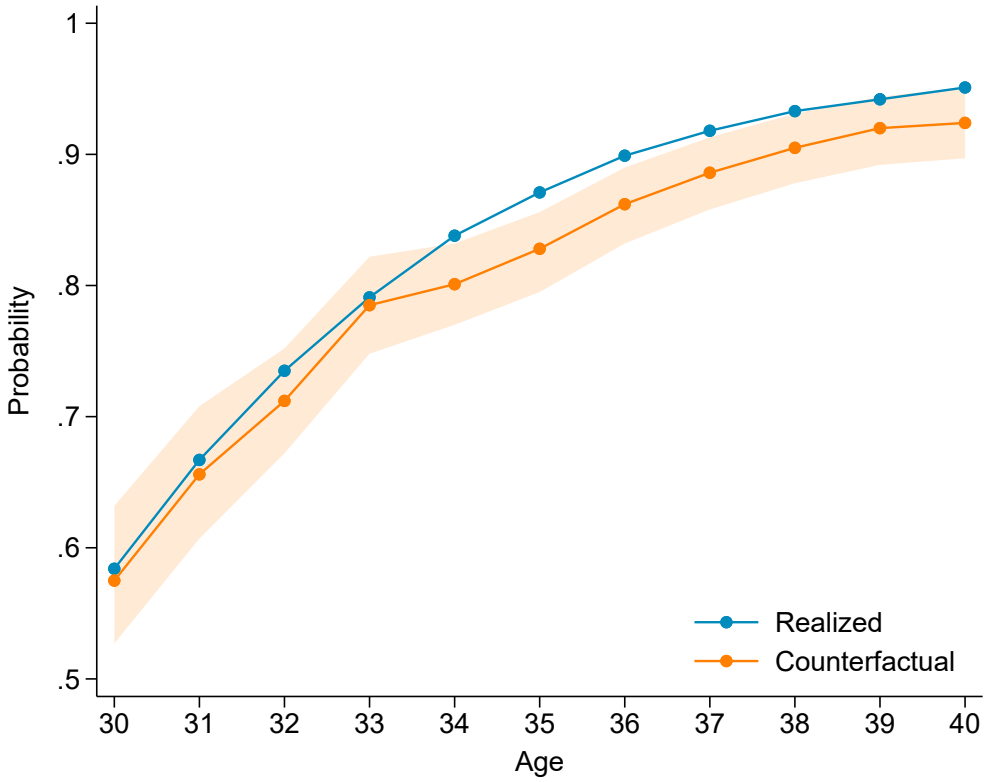
Note: The graph plots estimates of δ_ℓ from eq. (4), the effect of the first promotion of a former schoolmate to a managerial position on the average log hourly wage of the other members of the school cohort, ℓ years after the promotion, with 95% confidence intervals. Estimates follow Callaway and Sant'Anna (2021), using not-yet-treated groups as controls and long differences relative to the year before the promotion ($\ell = -1$). All specifications include fixed effects for region (NUTS3) $\times$ cohort $\times$ age and high-school track $\times$ age. Observations are school cohort $\times$ age cells, weighted by cell size. The average effect over the post-promotion period (ATT) is reported below the graph. Standard errors are clustered at the school level.

Figure 9: Schoolmate's Promotion and Wages, by Pay Premium of the Manager's Firm



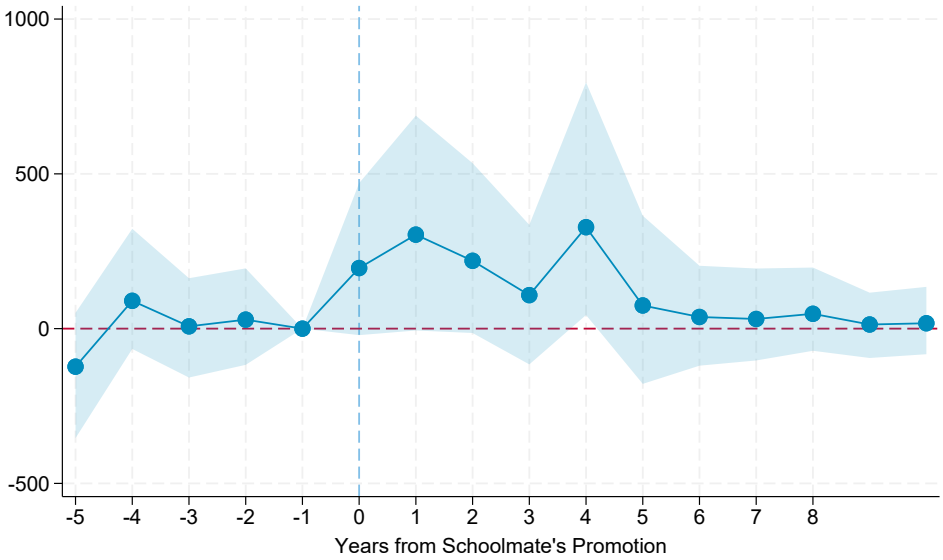
Note: The graph plots estimates of δ_ℓ from eq. (4), estimated separately for school cohorts whose first promoted manager works at a firm in the top 20% of the distribution of AKM firm effects (green diamonds) and for the other school cohorts (blue circles), with 95% confidence intervals. When more than one schoolmate is promoted at the same age, groups are classified by the highest firm effect among them. Not-yet-treated groups serve as controls in both samples. Specification as in Figure 8.

Figure 10: Former Schoolmate Is Manager: Realized and Counterfactual



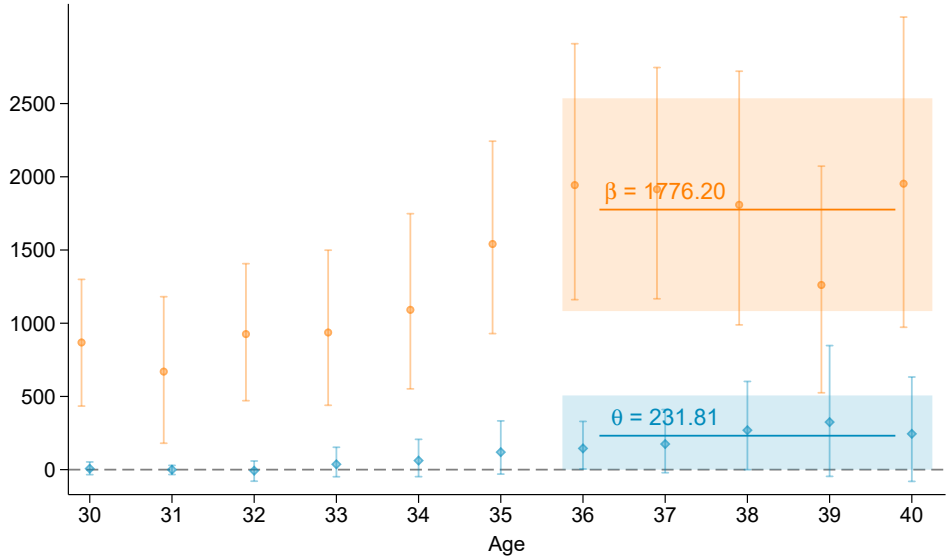
Note: The graph shows observed and counterfactual peer's promotion status for students exposed to high-SES peers ($P_i = r'$, top tercile of schoolmates' parental earnings). *Realized* (blue) is the share of students with at least one former schoolmate promoted to manager by each age. *Counterfactual* (orange) is the predicted share had the same students been exposed to low-SES peers (r), estimated subtracting the estimated effect of exposure, $\hat{\pi}^q$, from the realized share. The shaded area is the 90% confidence interval of the counterfactual from 1,000 bootstrap draws, clustered at the school level.

Figure 11: Effect of an Earlier Peer Promotion on Earnings



Note: The graph plots difference-in-differences estimates of the effect on annual earnings of experiencing the first peer promotion at age q rather than $q+1$, by years from the promotion, for students exposed to high-SES peers ($P_i = r'$). The control group consists of students whose first peer promotion occurs at $q+1$, balanced to match the treated group by cohort and school type. Estimates are aggregated across q, as a weighted average of the q-specific estimates using as weight π^q, the share of individuals experiencing an earlier promotion at each age q because of improved exposure. The shaded area is the 90% confidence interval from 1,000 bootstrap draws, clustered at the school level.

Figure 12: Total and Mediated Effect of Peer Exposure on Earnings

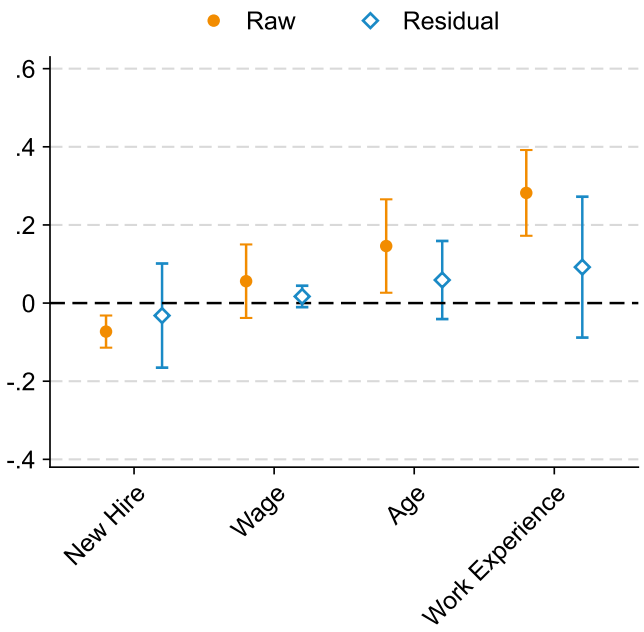


Note: The graph plots, by age, the total effect on annual earnings of exposure to high- rather than low-SES peers, β_a (orange circles), and the mediated effect operating through earlier peer promotions, θ_a (blue diamonds), from eq. (6), with 90% confidence intervals. Horizontal lines report the averages over ages 36–40, and shaded areas their 90% confidence intervals. Confidence intervals from 1,000 bootstrap draws, clustered at the school level.

A Further Balance Tests and Robustness

This appendix reports extra balance checks and estimates for alternative specifications of the model in Equation 1.

Figure A.1: Peer Quality and Teacher Characteristics



Note: The graph plots coefficients and 95% confidence intervals from separate regressions of the leave-one-out average of schoolmates' parental earnings on each teacher characteristic listed on the horizontal axis. Orange circles use school composition as observed in the data. Blue diamonds use school composition as the residual from school fixed effects, cohort fixed effects and school-specific time trends. Characteristics are standardized. Standard errors are clustered at the school level. Table A.1 reports the same estimates in table format.

Table A.1: Balance Test: Schoolmates' Parental Earnings and Teacher Characteristics

	Teachers		
	(1)	(2)	(3)
Wage	0.056	0.015	0.017
	(0.048)	(0.013)	(0.014)
	[0.248]	[0.258]	[0.238]
New Hire	-0.073	-0.005	-0.032
	(0.021)	(0.074)	(0.068)
	[0.001]	[0.949]	[0.639]
Age	0.146	0.016	0.059
	(0.061)	(0.039)	(0.051)
	[0.018]	[0.677]	[0.252]
Work Experience	0.282	0.008	0.092
	(0.056)	(0.078)	(0.092)
	[0.000]	[0.914]	[0.323]
School FEs		✓	✓
Cohort FEs		✓	✓
School T. Trends			✓

Note: Each cell reports the coefficient from a separate regression of schoolmates' parental earnings on the teacher characteristic indicated in the row. Standard errors are clustered at the school level and reported in parentheses, p-values are reported in square brackets. Column (1) measures school composition as observed in the data; columns (2) and (3) measure it as the residual from a linear model including the fixed effects indicated at the bottom of the table.

Table A.2: Peer Exposure, Predetermined Covariates and Predicted Earnings

	Earnings		Predicted Earnings
	(1)	(2)	(3)
Peers' Par. Earnings	0.071**	0.071**	0.000
	(0.029)	(0.030)	(0.001)
Predetermined Covariates		✓	
Observations	299,459	299,459	299,459

Note: The table reports OLS estimates of eq. (1). Column (1) reports the baseline specification. Column (2) adds as controls the predetermined student and parental characteristics used in the balance test of Table 4. In column (3) the dependent variable is earnings predicted by these characteristics. Earnings are measured in 2015 USD. All specifications include school fixed effects and school-specific time trends. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table A.3: Robustness to Alternative Fixed Effects and Trends

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: 2015 USD							
Par. Earnings	0.218***	0.219***	0.218***	0.218***	0.218***	0.250***	0.224***
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.05)	(0.04)
Peers' Par. Earnings	0.095***	0.108***	0.106***	0.091***	0.090***	0.063**	0.124***
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Panel B: Ranks							
Par. Earnings	0.146***	0.145***	0.146***	0.146***	0.146***	0.131***	0.132***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Peers' Par. Earnings	0.053***	0.056***	0.052***	0.041***	0.041***	0.054***	0.099***
	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Cohort FEs	✓	✓	✓	✓	✓	✓	✓
School FEs	✓	✓	✓	✓	✓	✓	✓
School T. Trends	✓	✓	✓	✓	✓		
Cohort × Munic. of Residence FEs		✓					
Cohort × Munic. of School FEs			✓				
School T. Trends (Quadratic)				✓			
School T. Trends (Cubic)					✓		
Pre/Post Moving Average (2 Cohorts)						✓	
Pre Moving Average (2 Cohorts)							✓
Observations	410,980	400,920	410,980	410,980	410,980	277,730	340,456

Note: The table reports OLS estimates of eq. (1) under alternative specifications, with fixed effects and trends indicated at the bottom of the table. The dependent variable is adult earnings, measured in 2015 USD in Panel A and in percentile ranks within birth cohort in Panel B; parental earnings and schoolmates' parental earnings are measured in the corresponding units. Column (1) reports the main specification. Columns (2) and (3) add cohort×municipality fixed effects, using the municipality of residence at age 16 and the municipality of the school, respectively. Columns (4) and (5) allow school-specific trends to be quadratic and cubic. Columns (6) and (7) replace school-specific trends with the moving average of schoolmates' parental earnings in the two preceding and two following cohorts of the same school (column 6), or in the two preceding cohorts only (column 7). All specifications control for gender and year of birth. Standard errors clustered at the school level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

B Estimating Sorting from Group Averages

Target. Let x_i be child i's parental rank and $g \equiv g(i)$ her group, with n_g children. The leave-one-out mean of her peers is

$$\bar{x}_{-i} = \frac{1}{n_g - 1} \sum_{j \in g,\, j \neq i} x_j.$$

The target is

$$\rho = \frac{\mathrm{Cov}(x_i, \bar{x}_{-i})}{\mathrm{Var}(x_i)}.$$

All moments are taken over children.

Step 1: Leave-one-out mean. Let $\mu_g = \mathbb{E}[x_i \mid g]$ and $e_i = x_i - \mu_g$, so that $\mathbb{E}[e_i \mid g] = 0$. Since the group total is $n_g \mu_g$, we have:

$$\bar{x}_{-i} = \frac{n_g \mu_g - x_i}{n_g - 1} = \mu_g - \frac{e_i}{n_g - 1}.$$

Step 2: Covariance. Because n_g and μ_g are constant within a group, $\mathbb{E}[x_i] = \mathbb{E}[\mu_g]$ and $\mathbb{E}[e_i \mid g] = 0$, we have:

$$\mathrm{Cov}(x_i, \mu_g) = \mathrm{Var}(\mu_g), \qquad \mathrm{Cov}\left(x_i, \frac{e_i}{n_g - 1}\right) = \mathbb{E}\left[\frac{\mathrm{Var}(x_i \mid g)}{n_g - 1}\right].$$

Hence:

$$\rho = \frac{\mathrm{Cov}(x_i, \mu_g) - \mathrm{Cov}\left(x_i, \frac{e_i}{n_g - 1}\right)}{\mathrm{Var}(x_i)} = \frac{\mathrm{Var}(\mu_g)}{\mathrm{Var}(x_i)} - \frac{1}{\mathrm{Var}(x_i)} \mathbb{E}\left[\frac{\mathrm{Var}(x_i \mid g)}{n_g - 1}\right].$$

Intuition: as the variance of group means approaches the total variance, little room is left for children to be different inside each group. The second term corrects for the fact that each child is excluded from her peer group by definition. The rest of the derivation is devoted to showing that it is small and bounded by the distribution of group size.

Let $\hat{\rho} = \mathrm{Var}(\mu_g)/\mathrm{Var}(x_i)$ be the between-group share of the variance. Then

$$\hat{\rho} = \rho + \frac{1}{\mathrm{Var}(x_i)} \mathbb{E}\left[\frac{\mathrm{Var}(x_i \mid g)}{n_g - 1}\right].$$

Observe that the error is larger the larger the within-group variance and the smaller the groups.

In our setting,[27] the distribution of group sizes is known and the variable of interest (x_i) is the percentile rank of parental earnings, which is uniformly distributed between 0 and 1. Therefore its variance within any group cannot exceed 1/4, the value reached when half of the group is at 0 and half at 1. Since $\mathrm{Var}(x_i) = 1/12$, we have:

$$0 \;\leq\; \hat{\rho} - \rho \;\leq\; \frac{1/4}{1/12}\,\mathbb{E}\left[\frac{1}{n_g - 1}\right] = 3\,\mathbb{E}\left[\frac{1}{n_g - 1}\right] = b.$$

Empirically, the bound $b = 3\,\mathbb{E}\left[\frac{1}{n_g - 1}\right]$ on the leave-one-out correction is small in all the settings considered. Table 7 compares the target parameter ρ, estimated directly from the leave-one-out regression, with the between-group share $\hat{\rho}$ and the implied bounds $[\hat{\rho} - b,\ \hat{\rho}]$ in two Danish settings where both can be computed. In both, the direct estimate lies within the bounds: $0.089 \in [0.059, 0.102]$ for school cohorts and $0.102 \in [0.091, 0.107]$ for municipalities. The bound is widest for school cohorts, the smallest groups, and narrows quickly with group size. U.S. Census tracts are larger still, so the bounds are the tightest of the three ($b = 0.005$), and ρ^{US} lies in $[0.277, 0.282]$. The between-group share is therefore a reliable estimate of the U.S. sorting coefficient, even though the target is not directly reported in publicly available data.

[27] Public data from the Opportunity Atlas (Chetty et al., 2026) report average parental earnings percentile rank and number of children by census tract. See Opportunity Atlas: Table 4 - All Outcomes by Census Tract, Race, Gender and Parental Income Percentile.

C Estimating Connected Hires with Dyadic Regression

This appendix describes how eq. (3) is estimated. The regression is defined on all pairs of a mover and a potential employer, in each period in which a move is observed. As individuals can move more than once, the baseline unit is a hire *event* ia: individual i being hired at age a. An observation is a pair $ia \times f$ of a hire event and a potential employer f, and I write $Z_{ia \times f}$ for any pair-level variable. The pooled sample contains more than 265 billion pairs, which I never construct.

The key insight is that, within a hire event, the identity of each firm is redundant. What matters is the connection status of each firm, along with the firm that hired the individual. In what follows, I describe how I obtain the exact OLS estimates of eq. (3) from data collapsed at this level.

Sample and definitions. I build a yearly panel of employment histories for the students in the main sample, at ages 25–39. The employer in each year is the plant of main employment, identified by its establishment identifier rather than by the identifier of the firm, so that plants of the same multi-plant firm are distinct potential employers. A *hire* occurs when an individual works at a plant different from the one in the previous year, starting either from another plant or from non-employment. The estimation sample consists of all hire events: one observation per individual and age at which a hire occurs.

Let $g = (s, c)$ denote a school cohort, and $g(i)$ the school cohort of individual i. Only tenured employees grant connections: an individual connects her schoolmates to a plant from her second year of employment there onward. For each school cohort g, plant f and year t, N^{C}_{gft} counts the tenured members of g employed at f in t. The almost-connection count N^{AC}_{gft} sums the same counts over the two adjacent cohorts of the same school, $(s, c-1)$ and $(s, c+1)$.

Timing and model. Time is measured by the age of the focal individual. Hire event ia takes place in calendar year $t_{ia} = c(i) + a - 16$, when $F_{t_{ia}}$ plants are active; these are the

potential employers. Connections are measured in that year:

$$C_{ia\times f} = \mathbb{1}\{N^C_{g(i)ft_{ia}} > 0\}, \qquad AC_{ia\times f} = \mathbb{1}\{N^C_{g(i)ft_{ia}} + N^{AC}_{g(i)ft_{ia}} > 0\}.$$

The estimated model is

$$H_{ia\times f} = \alpha_{ia} + \delta_1 C_{ia\times f} + \delta_2 AC_{ia\times f} + u_{ia\times f}, \qquad f = 1, \ldots, F_{t_{ia}}, \tag{7}$$

where $H_{ia\times f} = 1$ if f is the plant that hires i at age a. It is estimated separately for each age a and on all ages pooled.

Self-links. An individual is herself one of the tenured employees counted in $N^C_{g(i)ft_{ia}}$ if she already has tenure at the plant that hires her, for instance when she returns to a former employer. For every hire event, I therefore compute the connection status of the hiring plant after removing her from the count, so that no individual is connected to a plant through herself.

Collapsing the pairs. As $AC_{ia\times f} = 1$ implies $C_{ia\times f} = 1$, every pair connection status is characterized by $C_{ia\times f} + AC_{ia\times f} = k_{ia\times f} \in \{0, 1, 2\}$. Moreover, exactly one plant hires the individual, so $H_{ia\times f} = 1$ for one pair and 0 for all others. To properly collapse the pairs without losing information, the following weights are computed for each moving event ia. First, the hiring firm has weight one. Second, all other firms are grouped by their status k, and assigned weight equal to their number. Table 12 illustrates this, with n^k_{ia} being the number of firms with connection status k, except the hiring firm.

Table 12: Rows Representing One Hire Event ia

	Status k	$(C,\ AC)$	H	Weight w
Status of a plant				
Unconnected	0	(0, 0)		
Almost connected only	1	(0, 1)		
Connected	2	(1, 1)		
Rows				
1 Hiring plant	0, 1 or 2	by k	1	1
2 Other unconnected plants	0	(0, 0)	0	n_{ia}^0
3 Other almost connected only plants	1	(0, 1)	0	n_{ia}^1
4 Other connected plants	2	(1, 1)	0	n_{ia}^2

The counts n_{ia}^k come from collapsing N_{gft}^C and N_{gft}^{AC} to the school cohort–year level, so the size of the data grows with the number of hire events and of connected plants, not with the number of potential employers. On this collapsed data, I estimate eq. (7) by OLS with frequency weights w, with H divided by 100 so that coefficients are in percentage points.

Equivalence with the full regression. OLS with frequency weights solves the same normal equations as OLS on the data in which each row is repeated w times. Since rows are grouped only when they are identical in (ia, H, C, AC), the weighted regression reproduces the coefficients of the regression on all pairs exactly. The same holds for the standard errors. The cluster-robust variance depends on the data only through the within-cluster sums of $x_{ia\times f}\,\hat{u}_{ia\times f}$, where $x_{ia\times f} = (1, C_{ia\times f}, AC_{ia\times f})$, and through the number of observations, and frequency weights preserve both. Standard errors are clustered at the school level, the level at which school cohorts, and therefore connections, are assigned.

Single versus individual-specific intercepts. I estimate the model both with individual-specific intercepts α_{ia}, as in eq. (3), and with a single intercept α, as in eq. (7). The estimates of δ_1 and δ_2 do not change. This is expected: every hire event contributes exactly one hire, so individual intercepts mainly absorb differences in $F_{t_{ia}}$ across years. I therefore report the single-intercept estimates, and use $\hat{\alpha}$ to measure the baseline hiring probability at an almost-connected plant, $\hat{\alpha} + \hat{\delta}_2$, in the denominator of the Increase in P. Estimates with individual-specific intercepts are available upon request.

Reported quantities. The coefficient δ_1 is the difference in hiring probability between a connected plant and a plant with only almost connections, and δ_2 is the difference between a plant with only almost connections and an unconnected plant. The baseline hiring probability at an unconnected plant is close to $1/F_{t_{ia}}$, which is why the constant is small. I express the effect of a connection relative to the hiring probability at an almost-connected plant,

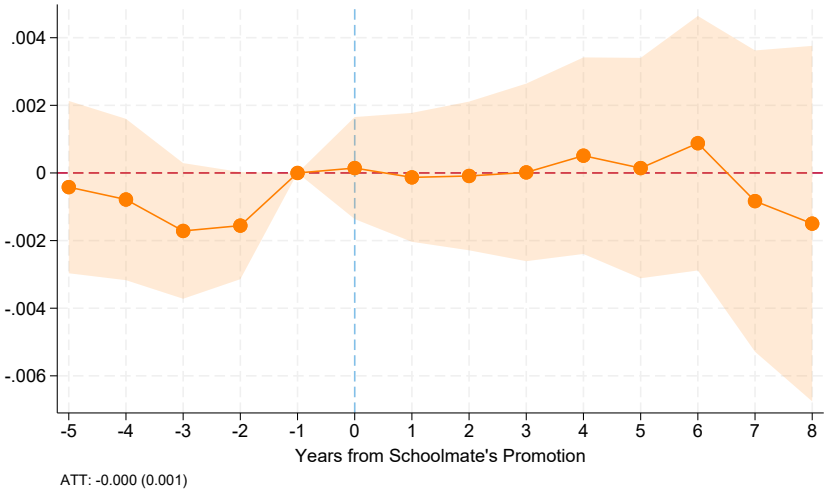
$$\text{Increase in } P = 100 \times \frac{\hat{\delta}_1}{\hat{\alpha} + \hat{\delta}_2},$$

with standard errors from the delta method.

Heterogeneity. Estimates by own background are obtained by running the regression separately for movers from the top tercile of parental earnings (rich movers) and for the others (poor movers). For hires at high-paying plants, the outcome is replaced by $H^G_{ia\times f}$, equal to one if f hires i at age a and f is in the top tercile of the AKM pay-premium distribution; the pairs and the weights are unchanged. To distinguish connections by the peer's background, the counts N^C_{gft} and N^{AC}_{gft} are computed separately for tenured schoolmates with parents in the top tercile of parental earnings (rich peers) and for the others (poor peers). Each pair then has one of nine statuses, combining its status with respect to rich and to poor peers, and the regression includes the four indicators $C^R_{ia\times f}$, $AC^R_{ia\times f}$, $C^P_{ia\times f}$ and $AC^P_{ia\times f}$, defined as above for each type of peer.

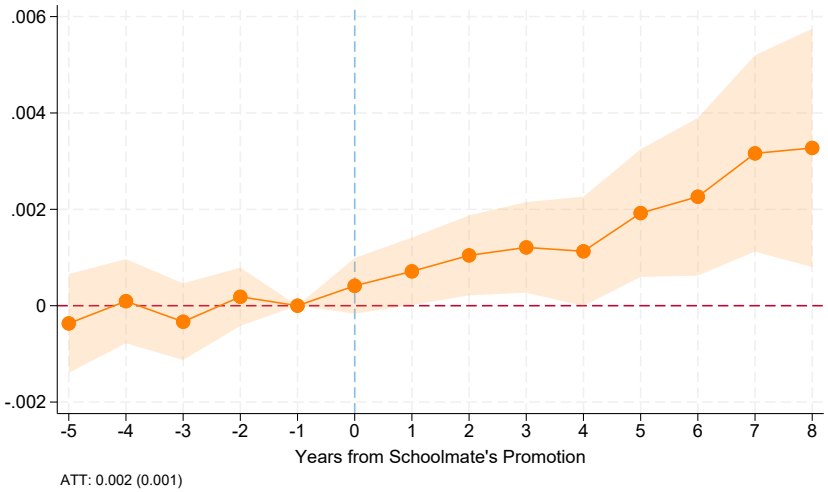
D Schoolmates' Promotions: Additional Results

Figure D.1: Schoolmate's Promotion and Employment



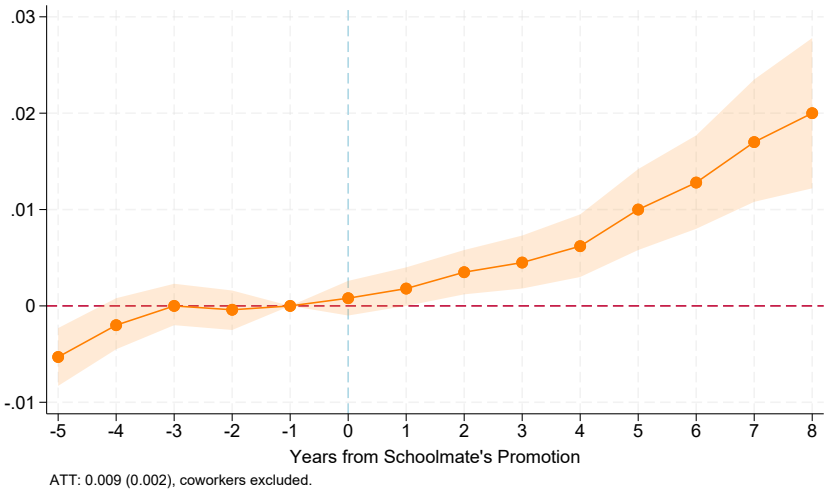
Note: The graph plots estimates of δ_ℓ from eq. (4) with the share of employed school-cohort members as outcome, with 95% confidence intervals. Specification as in Figure 8.

Figure D.2: Schoolmate's Promotion and the Employer's Pay Premium



Note: The graph plots estimates of δ_ℓ from eq. (4) with the average AKM pay premium of the employers of school-cohort members as outcome, with 95% confidence intervals. Specification as in Figure 8.

Figure D.3: Schoolmate's Promotion and Wages, Excluding the Manager's Coworkers



Note: The graph plots estimates of δ_ℓ from eq. (4) for log hourly wages, excluding from each school cohort the members who have worked at the same plant as the promoted manager, with 95% confidence intervals. The average effect over the post-promotion period (ATT) is reported below the graph. Specification as in Figure 8.

Table D.1: Average Effect on Wages, Alternative Estimators

	ATT	s.e.	95% C.I.
CS	0.009	0.002	[0.006, 0.013]
dCDH	0.014	0.003	[0.009, 0.020]
BJS	0.011	0.002	[0.007, 0.015]

Note: The table reports the average effect of a schoolmate's promotion on log hourly wages over the post-promotion period, with standard errors and 95% confidence intervals. CS: Callaway and Sant'Anna (2021), the baseline estimator. dCDH: de Chaisemartin and D'Haultfœuille (2024). BJS: imputation estimator of Borusyak et al. (2024). Standard errors are clustered at the school level.

E Mediation Analysis

This appendix presents the potential-outcomes framework underlying the mediation analysis in Section 6. The treatment is exposure to peers from different parental backgrounds, and the mediator is the timing of the first managerial promotion among an individual's former schoolmates. The framework extends Schenk (2025) to accommodate an ordered, event-time mediator and connect the paper's within-school exposure design to the promotion-timing design.

E.1 Setup and Target Parameter

Throughout, time is measured by the age of the individual. Let $Y_{i,a} \in \mathbb{R}$ denote the earnings of individual i at age $a \in \{a_0, \ldots, A\}$. Before a_0, the individual receives a time-invariant treatment $P_i \in \mathcal{P}$ and subsequently experiences an event-time mediator $S_i \in \mathcal{T} \equiv \{a_0, \ldots, A, \infty\}$. The value $S_i = \infty$ indicates that the event does not occur during the observation window. In the application, $P_i = r'$ denotes exposure in the top tercile of the leave-one-out distribution of schoolmate parental earnings and $P_i = r$ denotes exposure in the bottom tercile. The mediator S_i is the age at which the first former schoolmate is promoted to a managerial position.

Let $Y_{i,a}(p, s)$ be the potential outcome at age a under treatment p and mediator date s, and let $S_i(p)$ be the potential mediator date under treatment p. Only realized outcomes are observed: $Y_{i,a} = Y_{i,a}\big(P_i, S_i(P_i)\big)$, $S_i = S_i(P_i)$. For a change in peer parental background from r to r', define the average total, mediated, and direct effects on the treated as

$$\beta_a(r, r') \equiv \mathbb{E}[Y_{i,a}(r', S_i(r')) - Y_{i,a}(r, S_i(r)) \mid P_i = r'], \tag{8}$$

$$\theta_a(r, r') \equiv \mathbb{E}[Y_{i,a}(r', S_i(r')) - Y_{i,a}(r', S_i(r)) \mid P_i = r'], \tag{9}$$

$$\delta_a(r, r') \equiv \mathbb{E}[Y_{i,a}(r', S_i(r)) - Y_{i,a}(r, S_i(r)) \mid P_i = r']. \tag{10}$$

By construction, $\beta_a(r, r') = \theta_a(r, r') + \delta_a(r, r')$. The parameter of interest is $\theta_a(r, r')$, the component of the treatment effect operating through treatment-induced changes in peer-promotion timing.

E.2 Shift Decomposition

Define the effect of every one-year shift in promotion time at age $q \in \{a_0, \ldots, A\}$:

$$\Delta_{i,a}^{q} \equiv Y_{i,a}(r', q) - Y_{i,a}(r', q+1),$$

with the convention that $A + 1 = \infty$. For any two mediator dates k and k', the difference in potential outcomes can be written as the sum of the effects of crossing the intervening thresholds:

$$\begin{aligned} Y_{i,a}(r', k') - Y_{i,a}(r', k) &= \sum_{q=a_0}^{A} \Delta_{i,a}^{q} \mathbb{1}\{k' \le q < k\} - \sum_{q=a_0}^{A} \Delta_{i,a}^{q} \mathbb{1}\{k' > q \ge k\} \\ &= \sum_{q=a_0}^{A} \Delta_{i,a}^{q} \left(\mathbb{1}\{k' \le q < k\} - \mathbb{1}\{k' > q \ge k\}\right). \end{aligned}$$

Exposure to higher-SES peers may bring the first peer promotion forward for some individuals and delay it for others. Applying the shift decomposition with $k' = S_i(r')$ and $k = S_i(r)$ gives terms for crossings in both directions:

$$\theta_a(r, r') = \sum_{q=a_0}^{A} \left(\tau_a^{q,+}\, \pi^{q,+} - \tau_a^{q,-}\, \pi^{q,-}\right), \tag{11}$$

where

$$\tau_a^{q,+} = \mathbb{E}\left[\Delta_{i,a}^{q} \mid P_i = r',\ S_i(r') \le q < S_i(r)\right], \qquad \pi^{q,+} = \mathbb{P}[S_i(r') \le q < S_i(r) \mid P_i = r']$$

refer to individuals whose first peer promotion exposure brings forward across threshold q, and $\tau_a^{q,-}$ and $\pi^{q,-}$ are defined in the same way for individuals whose promotion it delays across q, with $S_i(r) \le q < S_i(r')$. Define the cumulative mediator indicator $M_i^q(p) = \mathbb{1}\{S_i(p) \le q\}$. Since $\mathbb{1}\{S_i(r') \le q < S_i(r)\} - \mathbb{1}\{S_i(r) \le q < S_i(r')\} = M_i^q(r') - M_i^q(r)$, the net share of threshold crossers is

$$\pi_c^q(r, r') \equiv \pi^{q,+} - \pi^{q,-} = \mathbb{E}[M_i^q(r') - M_i^q(r) \mid P_i = r'].$$

E.3 Identifying the Crosser Share

As defined above, the net crosser share is

$$\pi_c^q(r, r') = \mathbb{E}[M_i^q(r') - M_i^q(r) \mid P_i = r'] \,. \tag{12}$$

Identification of the crosser share therefore reduces to identification of the treatment effect on the probability that a peer promotion has occurred by threshold q.

As in eq. (1), let X_i denote the parental earnings of individual i and $\bar{X}_{-i}$ the leave-one-out average parental earnings of her schoolmates. The discrete treatment regimes r and r' correspond to the bottom and top terciles of $\bar{X}_{-i}$. Let $\bar{x}_p = \mathbb{E}[\bar{X}_{-i} \mid P_i = p]$ be the mean exposure within regime p, and let $M_i^{*q}(\bar{x}_{-i})$ be the potential cumulative mediator under continuous exposure $\bar{X}_{-i} = \bar{x}_{-i}$. Finally, let H_i collect the constant and the controls of eq. (1): own parental earnings X_i, the predetermined characteristics W_i, school fixed effects and school-specific cohort trends.

Assumption A1: Linear-in-Means Model of Peer Exposure. For every threshold q and exposure level $\bar{x}_{-i}$ in the support of $\bar{X}_{-i}$,

$$\mathbb{E}[M_i^{*q}(\bar{x}_{-i}) \mid \alpha_i^q] = \alpha_i^q + \gamma^q \bar{x}_{-i}, \qquad \alpha_i^q = H_i' b^q + e_i^q.$$

Exposure shifts the potential mediator linearly and homogeneously, and the individual-specific level α_i^q has a component explained by the controls of eq. (1) and a residual component e_i^q. The first part of the condition yields

$$\pi_c^q(r, r') = \gamma^q(\bar{x}_{r'} - \bar{x}_r) \equiv \gamma^q \Delta_{\bar{x}}. \tag{13}$$

Since the observed mediator is $M_i^q = M_i^{*q}(\bar{X}_{-i})$, the second part gives

$$M_i^q = H_i' b^q + \gamma^q \bar{X}_{-i} + u_i^q, \tag{14}$$

which is eq. (1) with the cumulative mediator in place of earnings, where $u_i^q = e_i^q + M_i^q - \mathbb{E}[M_i^{*q}(\bar{X}_{-i}) \mid \alpha_i^q, \bar{X}_{-i}]$ collects the residual heterogeneity and the deviation of the realized mediator from its conditional mean.

Assumption A2: Quasi-Random Fluctuations in Cohort Composition. Let $\bar{X}_{-i} = H_i' b^X + Z_i$ be the population projection of exposure on the controls, so that Z_i is the residual variation in school composition around school-specific trends. For every threshold q,

$$\mathbb{E}[Z_i u_i^q] = 0.$$

This is the identifying assumption of eq. (1) applied to the mediator: conditional on school-specific trends and controls, residual variation in schoolmates' parental earnings is unrelated to residual determinants of peer-promotion timing. Under A1 and A2, OLS on eq. (14) identifies γ^q, and

$$\widehat{\pi}_c^q(r, r') = \widehat{\gamma}^q \Delta_{\bar{x}}.$$

E.4 Identifying the Effect of Earlier Peer Promotion

Let $\Delta Y_i^{a,q} = Y_{i,a} - Y_{i,q-1}$ and let K_i contain predetermined covariates used for adjustment.

Assumption A3: Promotion-Timing Design. For every threshold q and age $a \geq q$:

(a) *No anticipation.* $Y_{i,b}(r', s) = Y_{i,b}(r', \infty)$ for every s and every age $b < s$.

(b) *Parallel trends under a one-year delay.*

$$\begin{aligned}&\mathbb{E}[Y_{i,a}(r', q+1) - Y_{i,q-1}(r', q+1) \mid P_i = r',\, S_i = q,\, K_i] \\ &\quad = \mathbb{E}[Y_{i,a}(r', q+1) - Y_{i,q-1}(r', q+1) \mid P_i = r',\, S_i = q+1,\, K_i]\,.\end{aligned}$$

(c) *Overlap.* $\mathbb{P}[S_i = q+1 \mid P_i = r',\, S_i \in \{q, q+1\},\, K_i] > 0$ over the support of K_i among individuals with $S_i = q$.

Condition (a) implies $\Delta_{i,a}^{q} = 0$ for every threshold $q > a$: a peer promotion that has not yet occurred by age a does not affect earnings at that age. The terms with $q > a$ in eq. (11) therefore vanish.

Let $\omega(K_i)$ denote the weights that make the distribution of K_i among individuals with $S_i = q + 1$ equal to that among individuals with $S_i = q$, which exist under (c). Within the high-exposure regime, observed earnings are $Y_{i,a} = Y_{i,a}(r', S_i)$. Hence

$$\begin{aligned}
&\mathbb{E}[\Delta Y_i^{a,q} \mid P_i = r', S_i = q] - \mathbb{E}[\omega(K_i)\, \Delta Y_i^{a,q} \mid P_i = r', S_i = q+1] \\
&\quad = \mathbb{E}\big[\big(Y_{i,a}(r', q) - Y_{i,q-1}(r', q)\big) - \big(Y_{i,a}(r', q+1) - Y_{i,q-1}(r', q+1)\big) \mid P_i = r', S_i = q\big] \\
&\quad = \mathbb{E}[Y_{i,a}(r', q) - Y_{i,a}(r', q+1) \mid P_i = r', S_i = q] = \tau_{a,\mathrm{DiD}}^{q},
\end{aligned}$$

where the first equality uses (b) integrated over the distribution of K_i among individuals with $S_i = q$, and the second uses (a), which gives $Y_{i,q-1}(r', q) = Y_{i,q-1}(r', q+1)$. The adjusted difference-in-differences contrast $\tau_{a,\mathrm{DiD}}^{q}$ therefore identifies the effect of a promotion at age q rather than $q + 1$ for individuals whose first peer promotion occurs at age q. The target parameter in eq. (11), however, requires this effect for threshold crossers, in both directions.

Assumption A4: Effect Transportability to Threshold Crossers. For every q and a,

$$\begin{aligned}
\mathbb{E}\big[\Delta_{i,a}^{q} \mid P_i = r',\ S_i(r') = q\big] &= \mathbb{E}\big[\Delta_{i,a}^{q} \mid P_i = r',\ S_i(r') \le q < S_i(r)\big] \\
&= \mathbb{E}\big[\Delta_{i,a}^{q} \mid P_i = r',\ S_i(r) \le q < S_i(r')\big]\,.
\end{aligned}$$

The effect of a promotion at q rather than $q + 1$ is the same, on average, for individuals whose first peer promotion occurs at q and those whose promotion is moved across q in either direction by peer exposure. The first equality transports the difference-in-differences estimand to threshold crossers. The second extends it to crossings in the opposite direction, so that no restriction on the direction of the effect of exposure on promotion timing is needed.

Under A4, $\tau^q_{a,\mathrm{DiD}} = \tau^{q,+}_a = \tau^{q,-}_a$, and, combining with A3(a),

$$\theta_a(r, r') = \sum_{q=a_0}^{a} \pi^q_c(r, r')\, \tau^q_{a,\mathrm{DiD}}.$$

E.5 Estimation and Inference

Sample. The sample is the main sample of the paper. A peer promotion is the first promotion to a managerial position among the former schoolmates of the same school cohort, at a private-sector firm with at least 25 employees, excluding the public sector and education, as in section 5. The analysis covers ages 30 to 40.

Crosser share. For each threshold q, γ^q is estimated by OLS on eq. (14). By the Frisch–Waugh–Lovell theorem, this is equivalent to

$$\widetilde{M}^q_i = \gamma^q \widetilde{X}_{-i} + u^q_i, \tag{15}$$

where tildes denote residuals after projection on H_i, so that $\widetilde{X}_{-i} = Z_i$. The exposure gap $\Delta_{\bar{x}}$ is the difference in the sample means of $\bar{X}_{-i}$ between the top and bottom terciles, computed once on the full sample.

Effect of an earlier promotion. For each threshold q, $\tau^q_{a,\mathrm{DiD}}$ is estimated among individuals in the high-exposure regime by the reweighted difference-in-differences contrast above, comparing $S_i = q$ with $S_i = q + 1$. The balancing covariates K_i are the full interaction of enrollment cohort and school type, so that $\omega(K_i)$ is the ratio of the shares of the two groups in each cell.

Mediated effect. The mediated effects at each age are

$$\widehat{\theta}_a(r, r') = \sum_{q=a_0}^{a} (\widehat{\gamma}^q \Delta_{\bar{x}})\, \widehat{\tau}^q_{a,\mathrm{DiD}}. \tag{16}$$

Inference. Inference is based on 1,000 draws of a Bayesian bootstrap (Rubin, 1981) at the school level. In each draw, every school receives a weight drawn independently from an

exponential distribution with mean one, shared by all its students, to account for clustering at the school level. The estimation of $\widehat{\gamma}^q$, the balancing weights and $\widehat{\tau}^q_{a,\text{DiD}}$ is then repeated with these weights, while the exposure gap $\Delta_{\bar{x}}$ is held fixed at its full-sample value. This preserves the covariance between the components of $\widehat{\theta}_a$ across thresholds. Reported standard errors are the standard deviation of the bootstrap distribution, and 90% confidence intervals are its 5th and 95th percentiles.

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